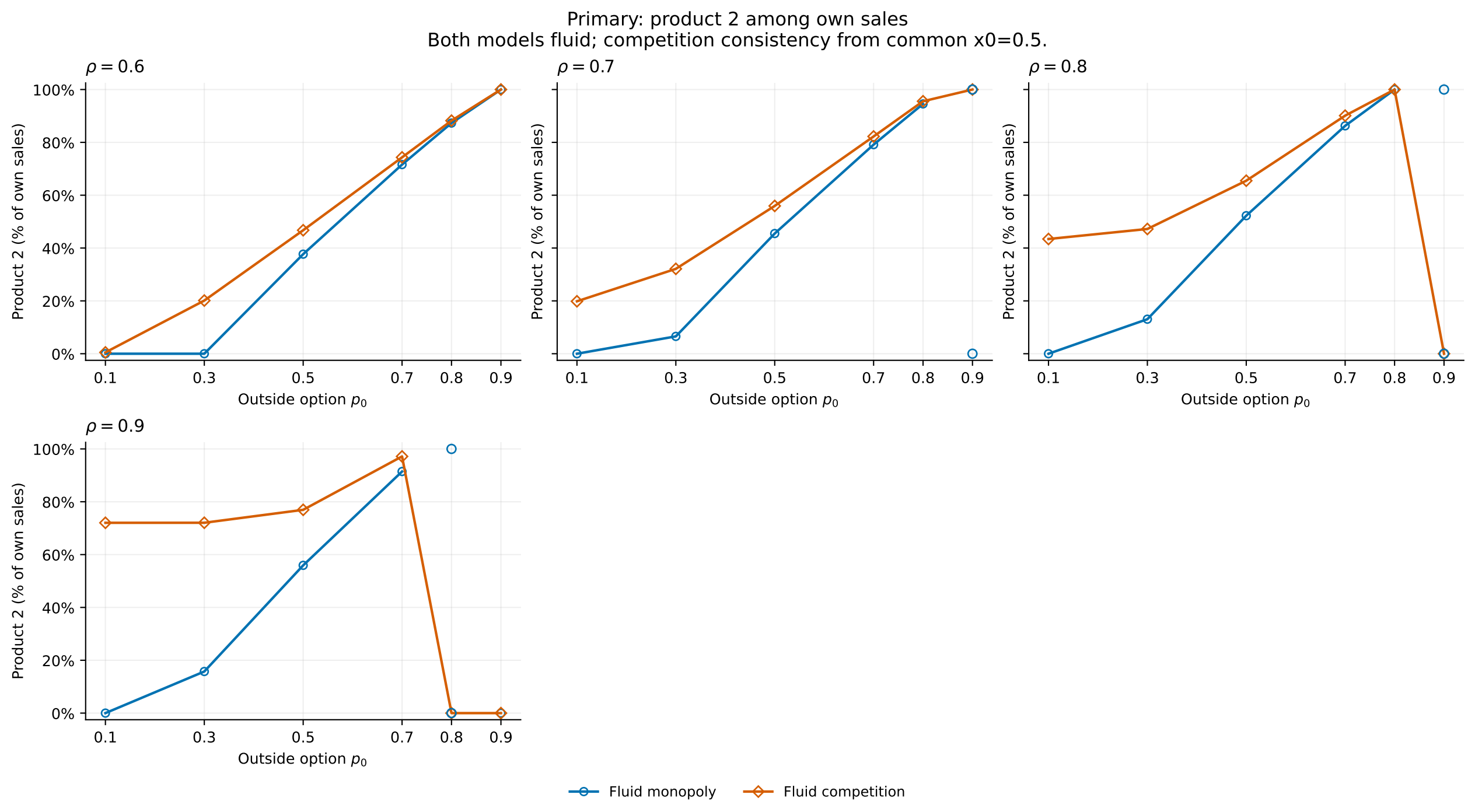
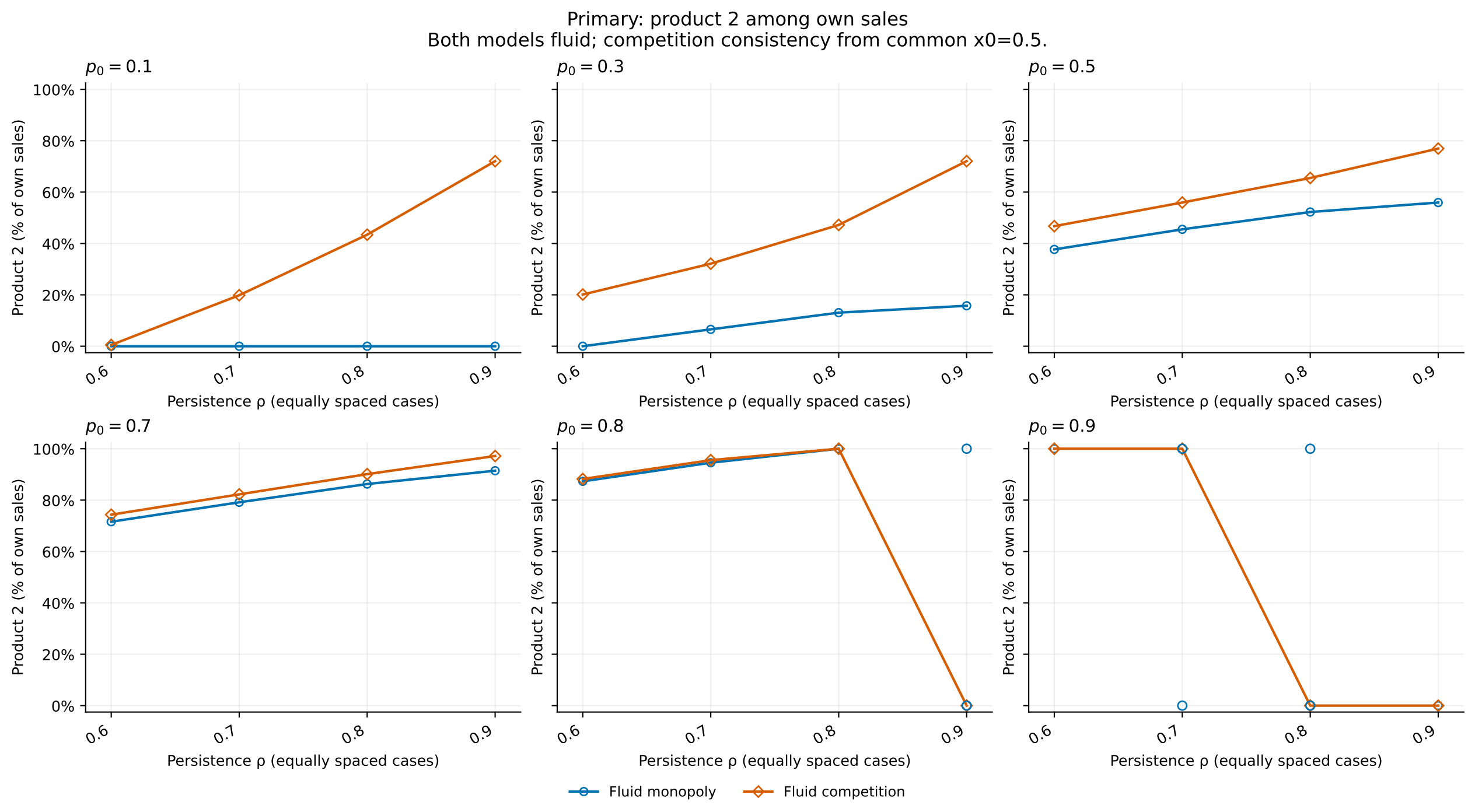
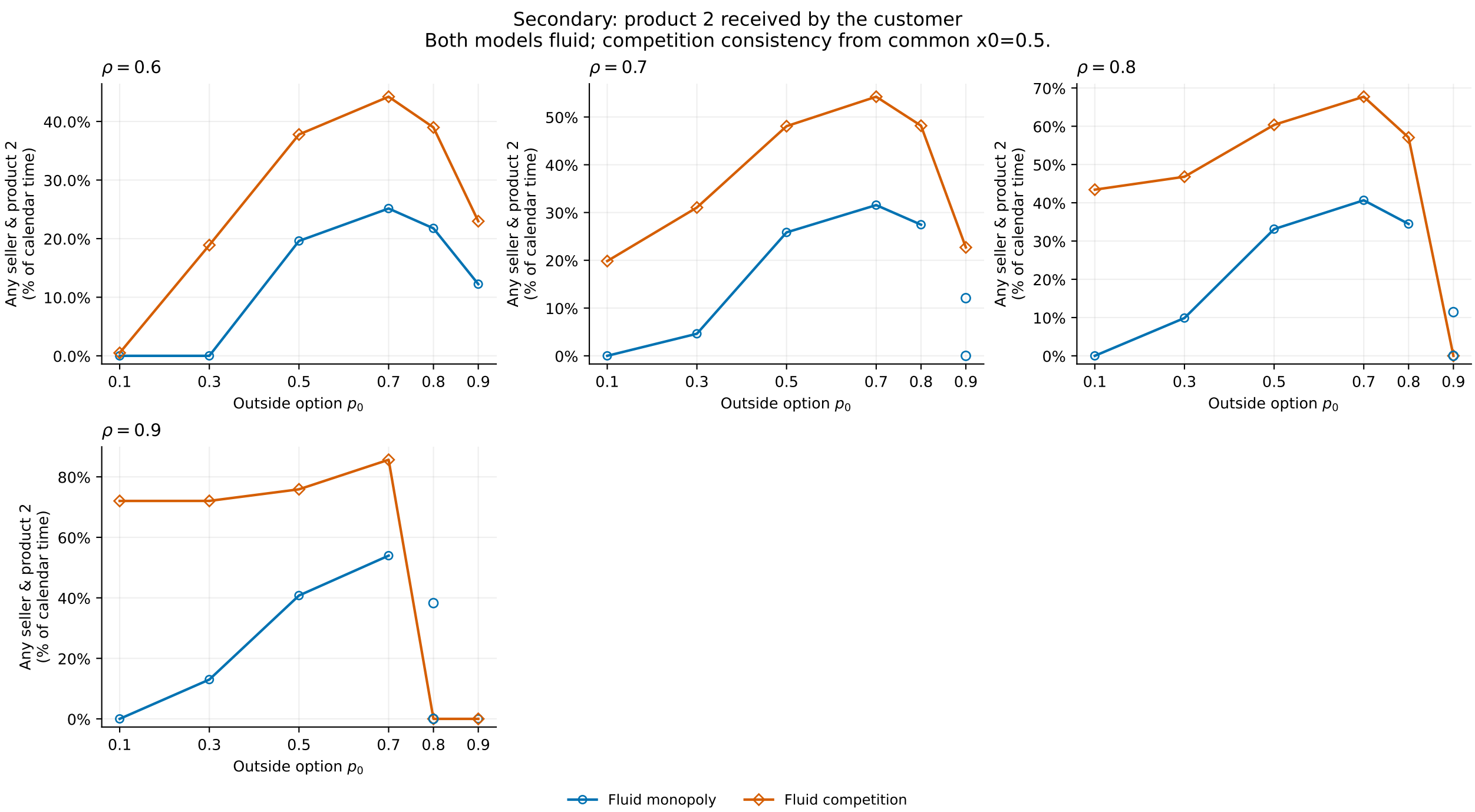
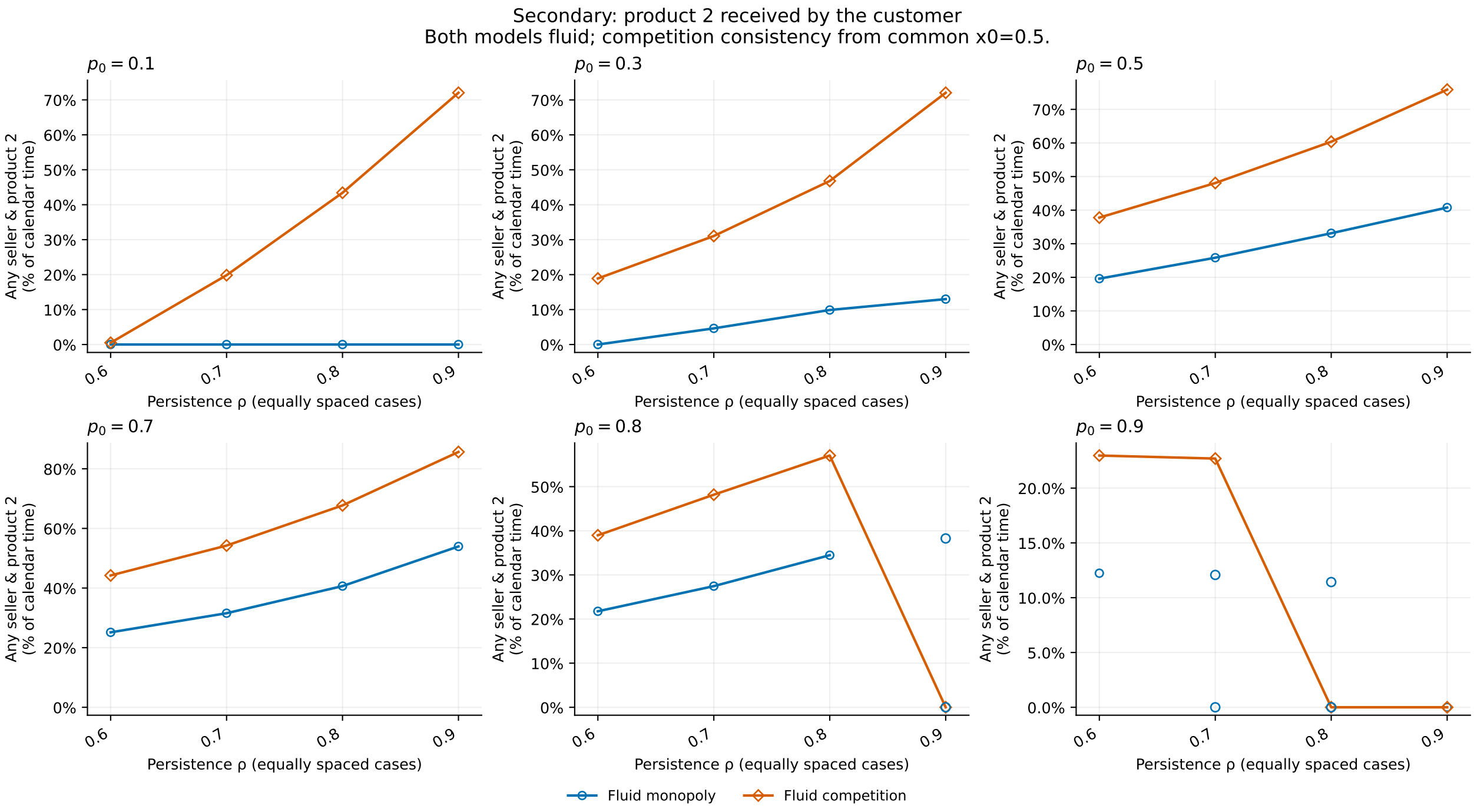
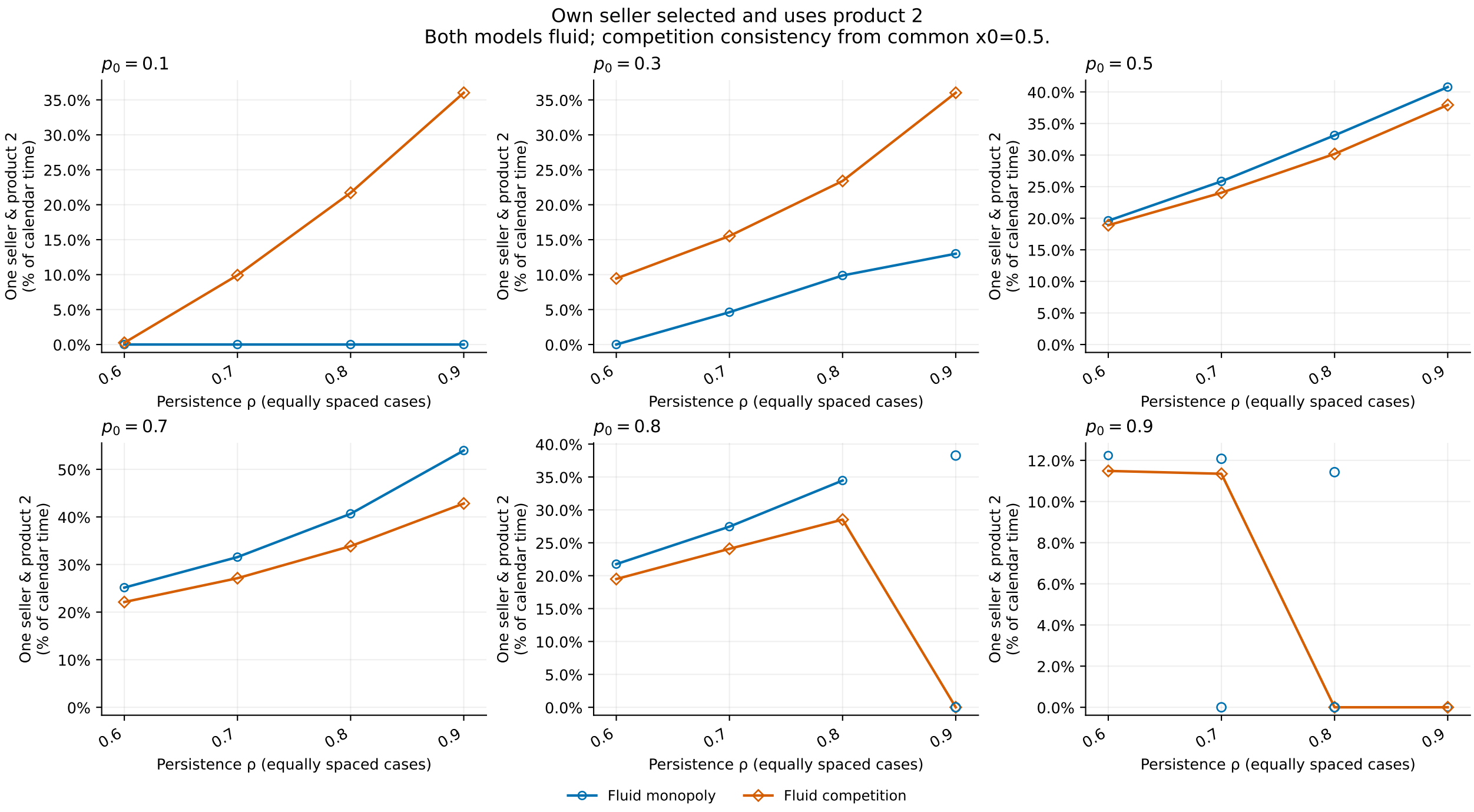
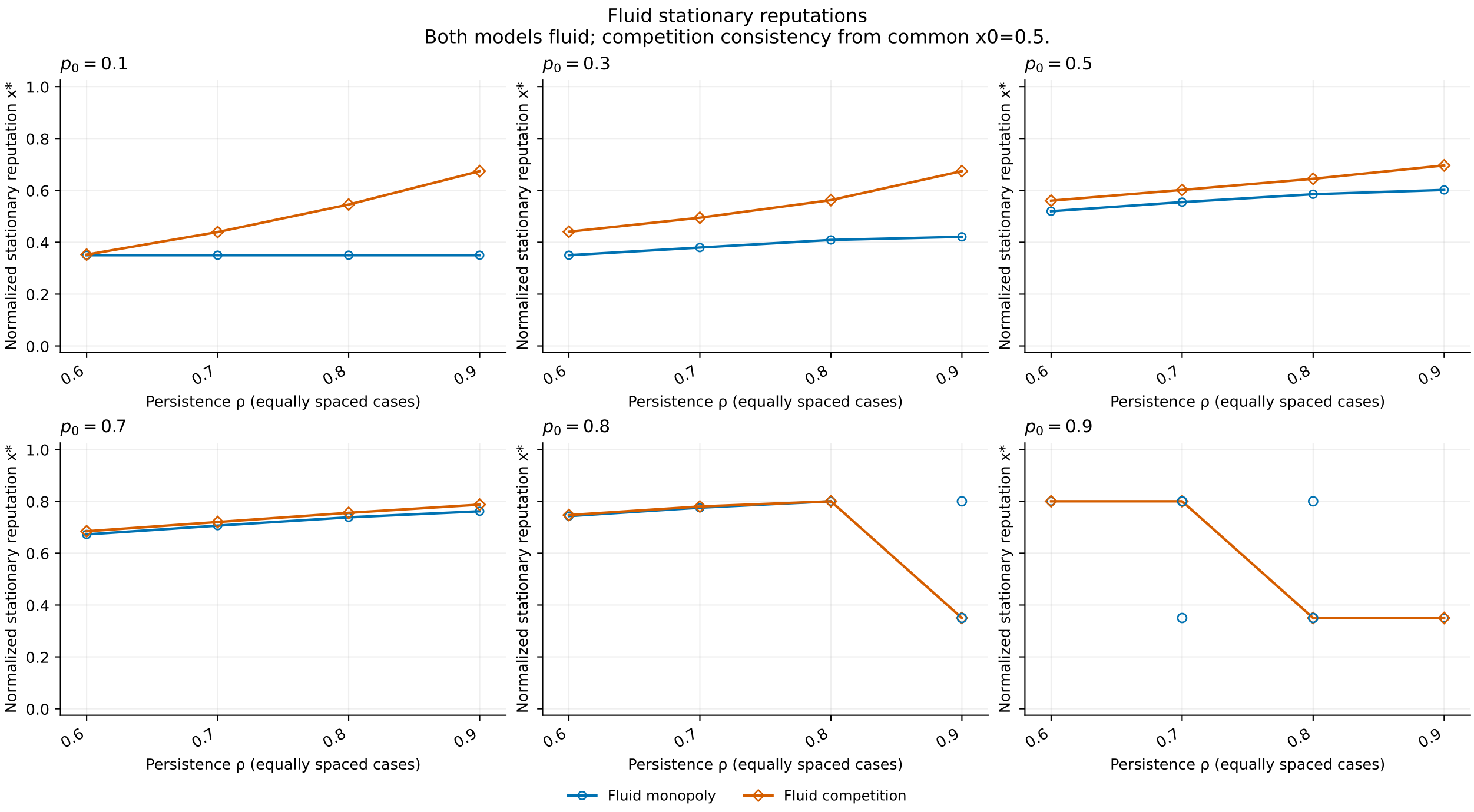

Primary: product 2 among own sales
Both models fluid; competition consistency from common x0=0.5.
ρ = 0.6
ρ = 0.7
ρ = 0.8
ρ = 0.9
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.1
0.3
0.5
0.7
0.8
0.9
Outside option p_0
Fluid monopoly
Fluid competition

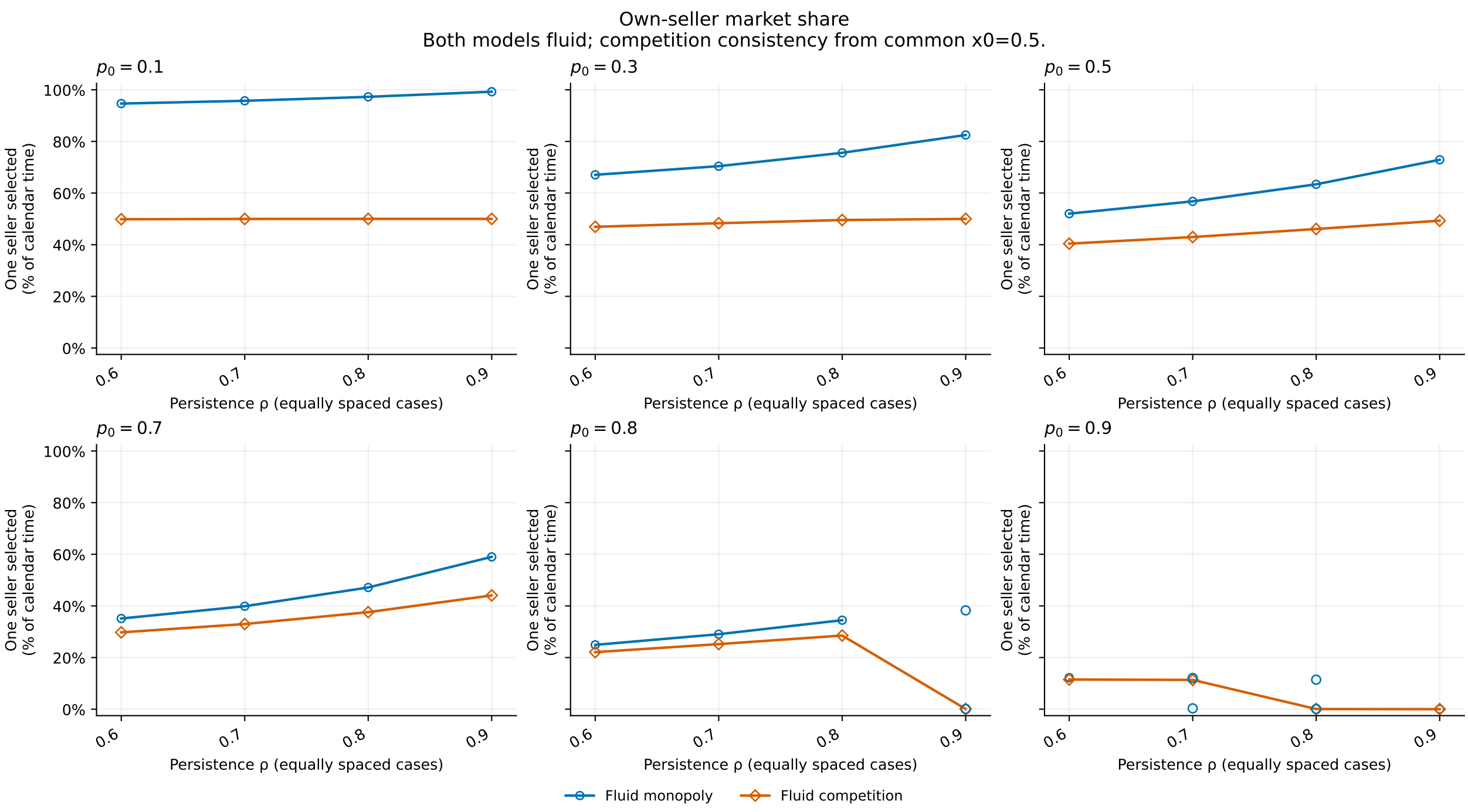

Primary: product 2 among own sales
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

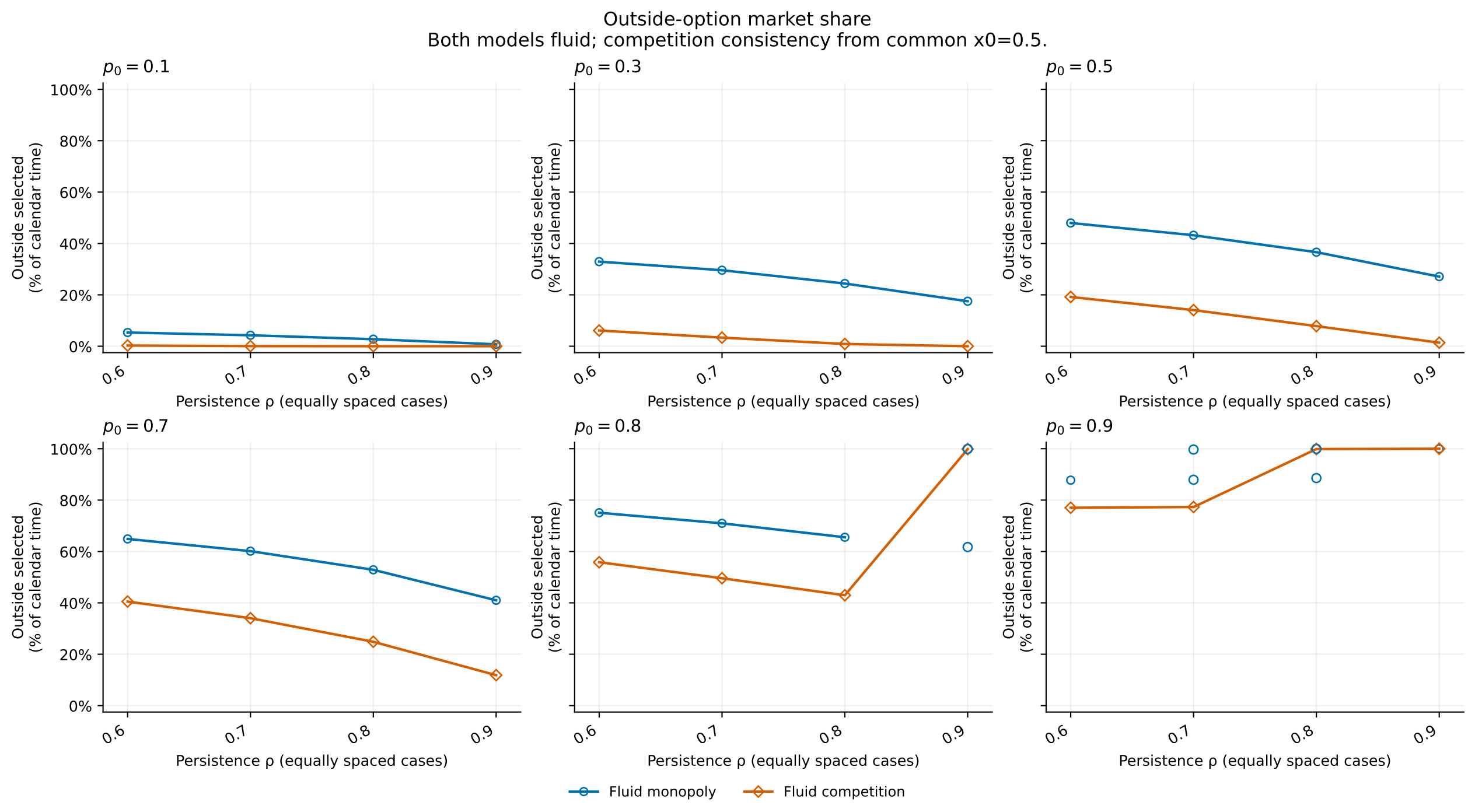

Secondary: product 2 received by the customer
Both models fluid; competition consistency from common x0=0.5.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Any seller & product 2
(% of calendar time)
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
Fluid monopoly
Fluid competition

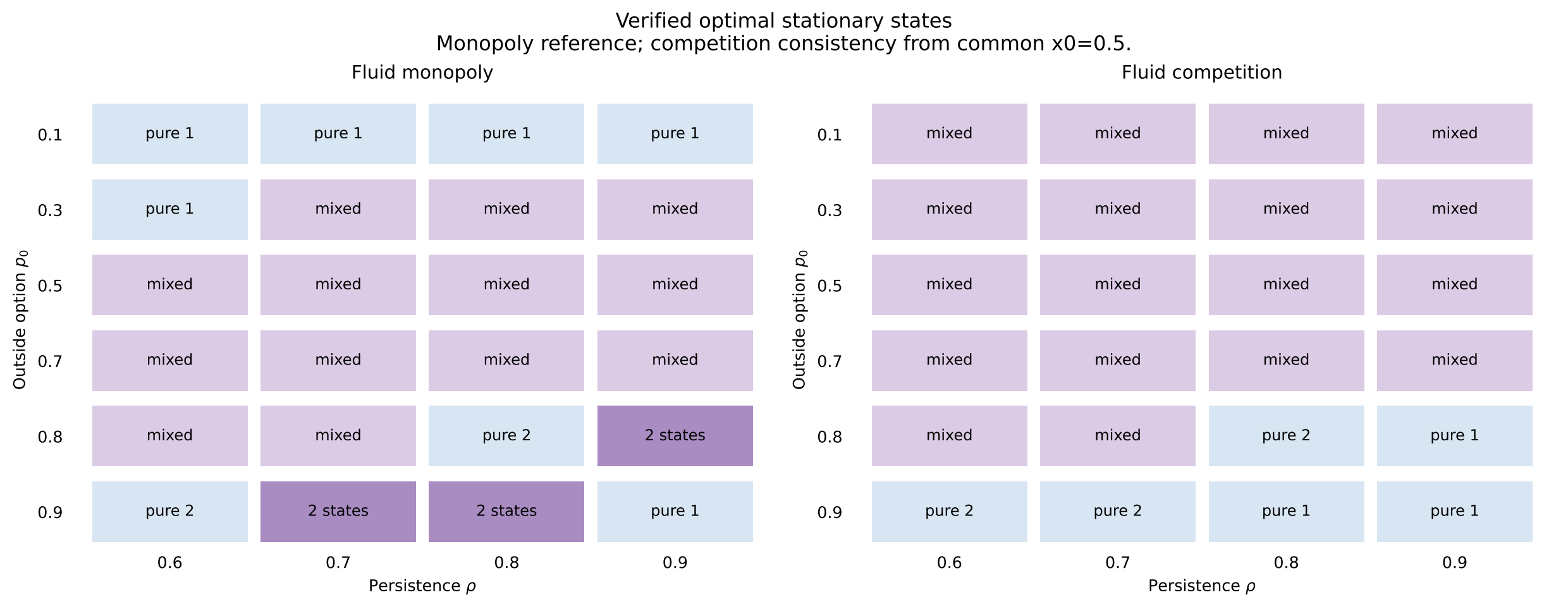

Secondary: product 2 received by the customer
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2
(% of calendar time)
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

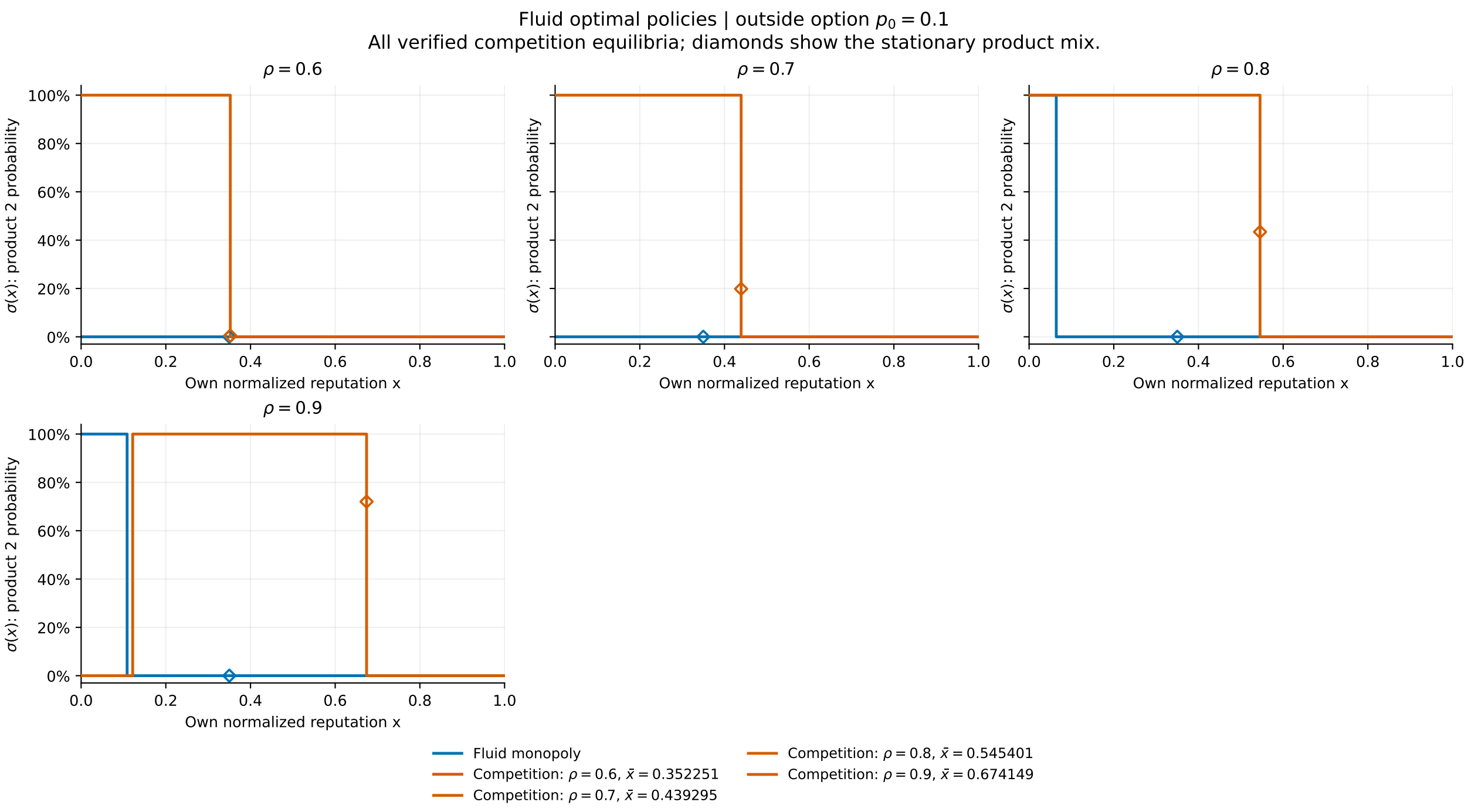

Own seller selected and uses product 2
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2
(% of calendar time)
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

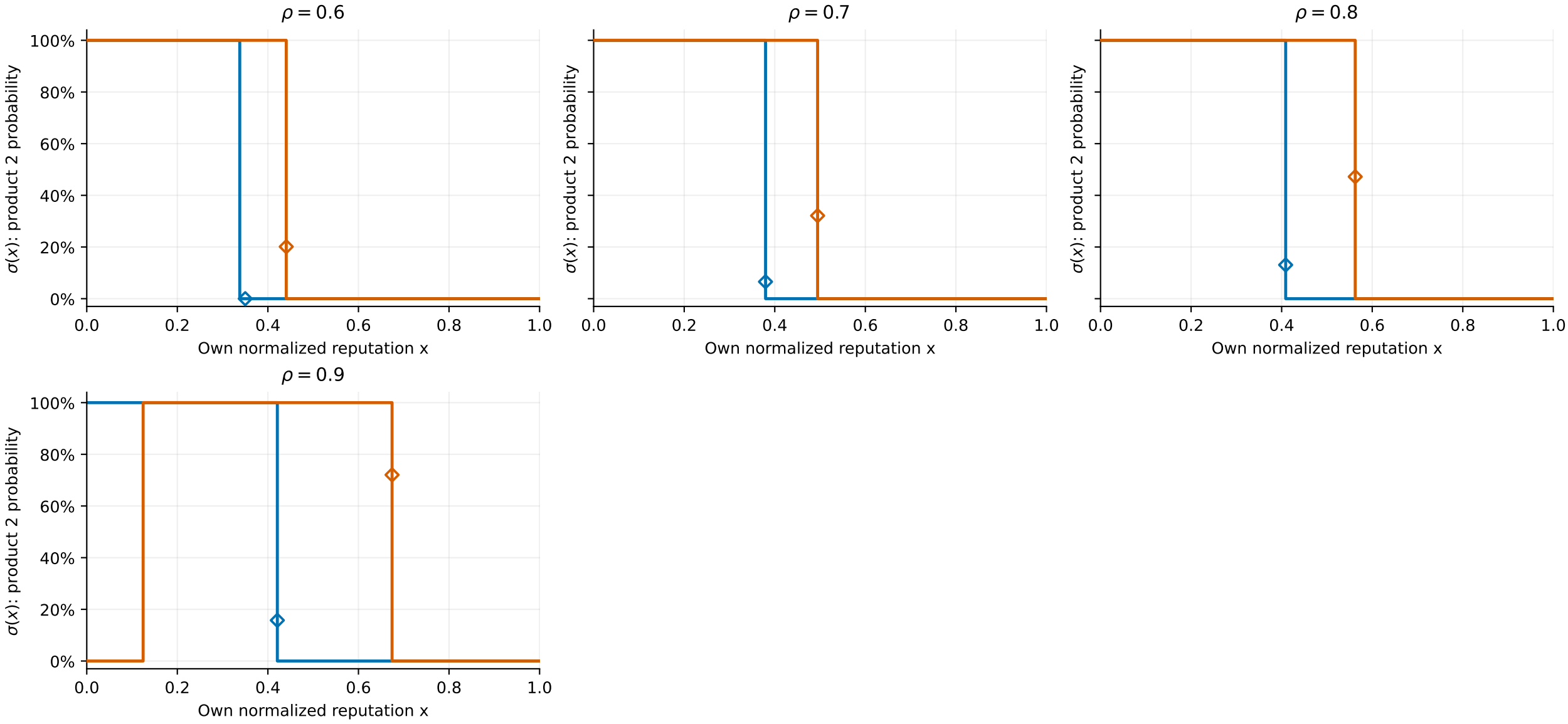

Fluid stationary reputations
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Normalized stationary reputation x*
Persistence ρ (equally spaced cases)
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.7
0.8
0.9
Fluid monopoly
Fluid competition

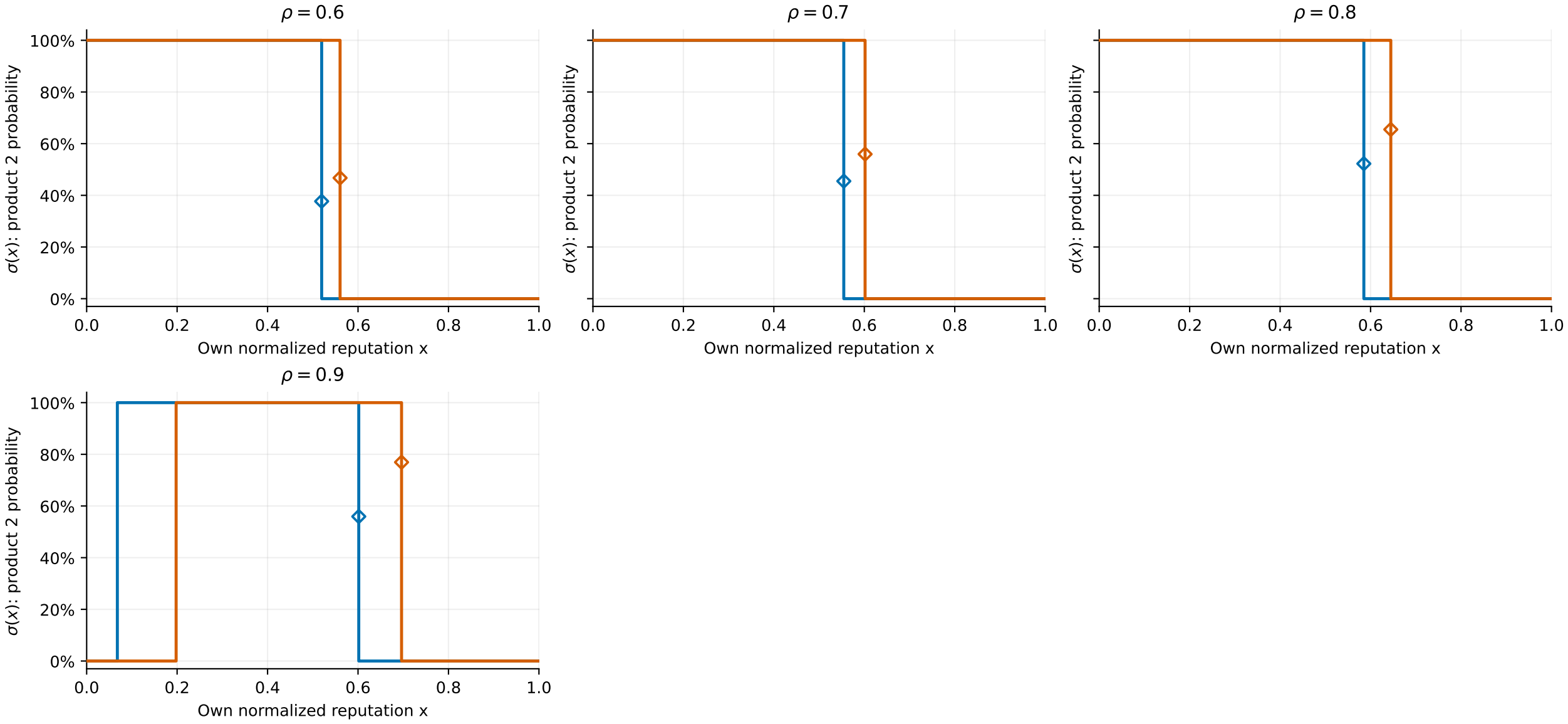

Own-seller market share
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller selected
(% of calendar time)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

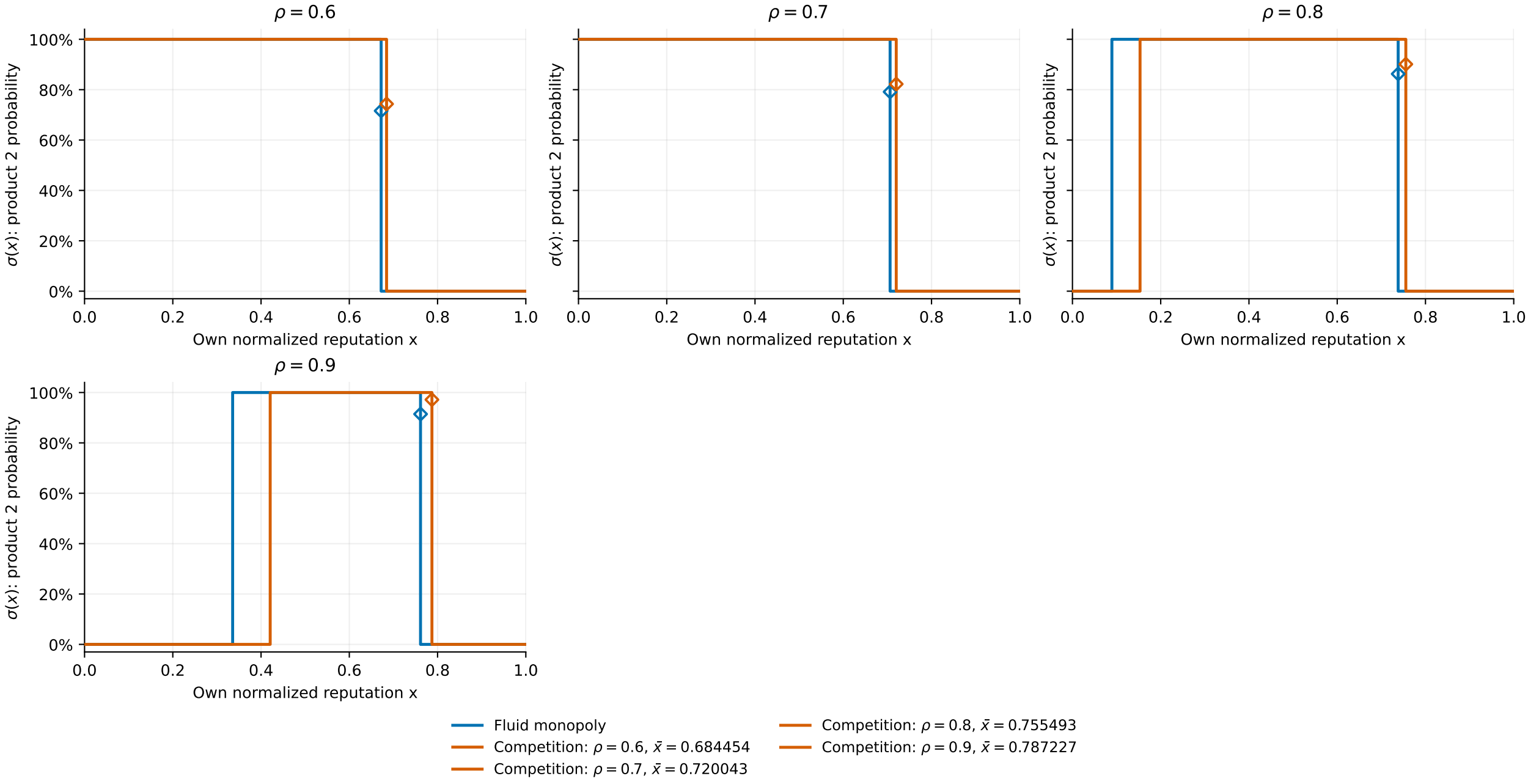

Outside-option market share
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Outside selected (% of calendar time)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

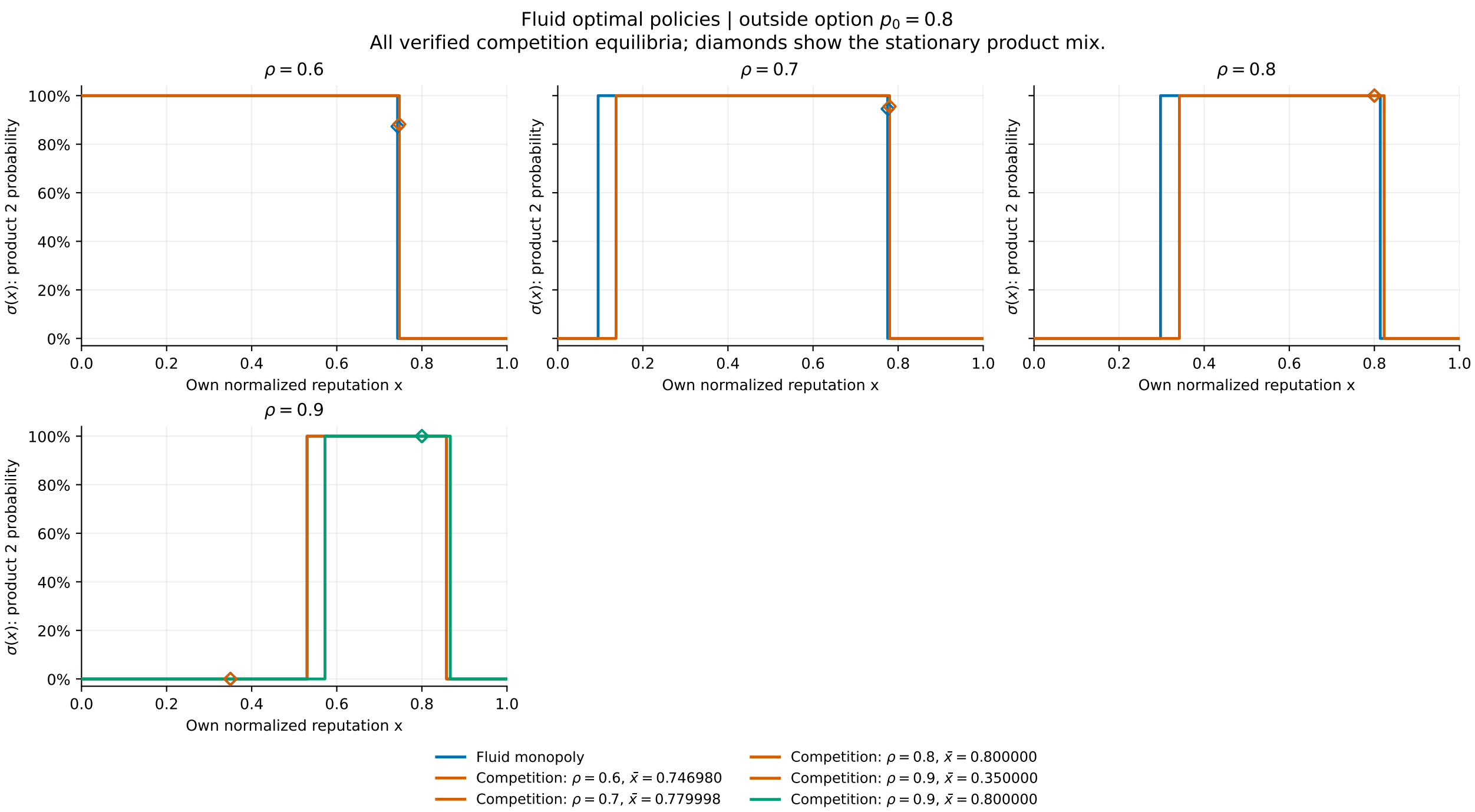

Verified optimal stationary states
Monopoly reference; competition consistency from common x0=0.5.
Fluid monopoly
Fluid competition
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
pure 1
pure 2
mixed
2 states
0.6
0.7
0.8
0.9
Persistence ρ

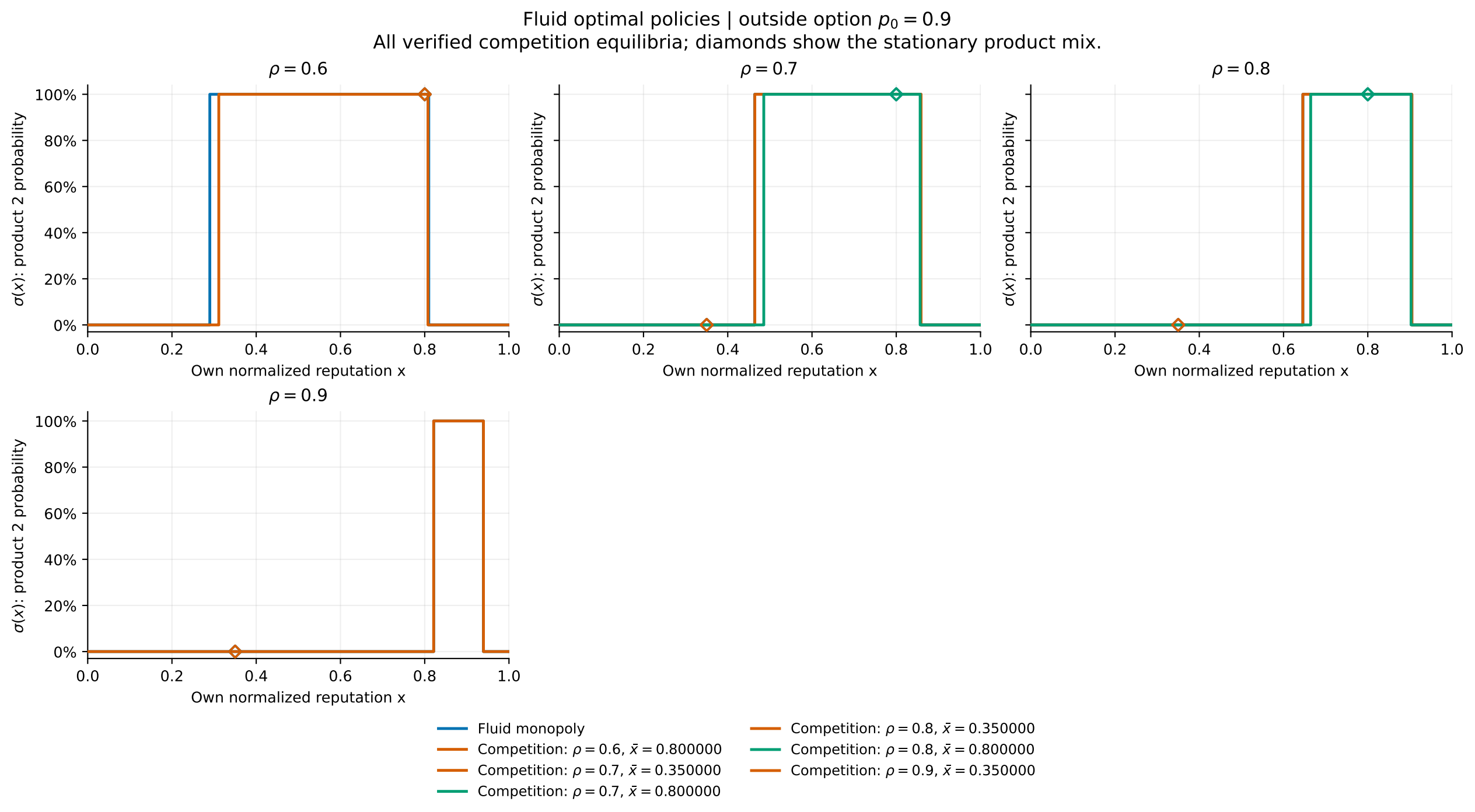

Fluid optimal policies | outside option $p_0 = 0.1$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.352251$
Competition: $\rho = 0.7$, $\bar{x} = 0.439295$
Competition: $\rho = 0.8$, $\bar{x} = 0.545401$
Competition: $\rho = 0.9$, $\bar{x} = 0.674149$

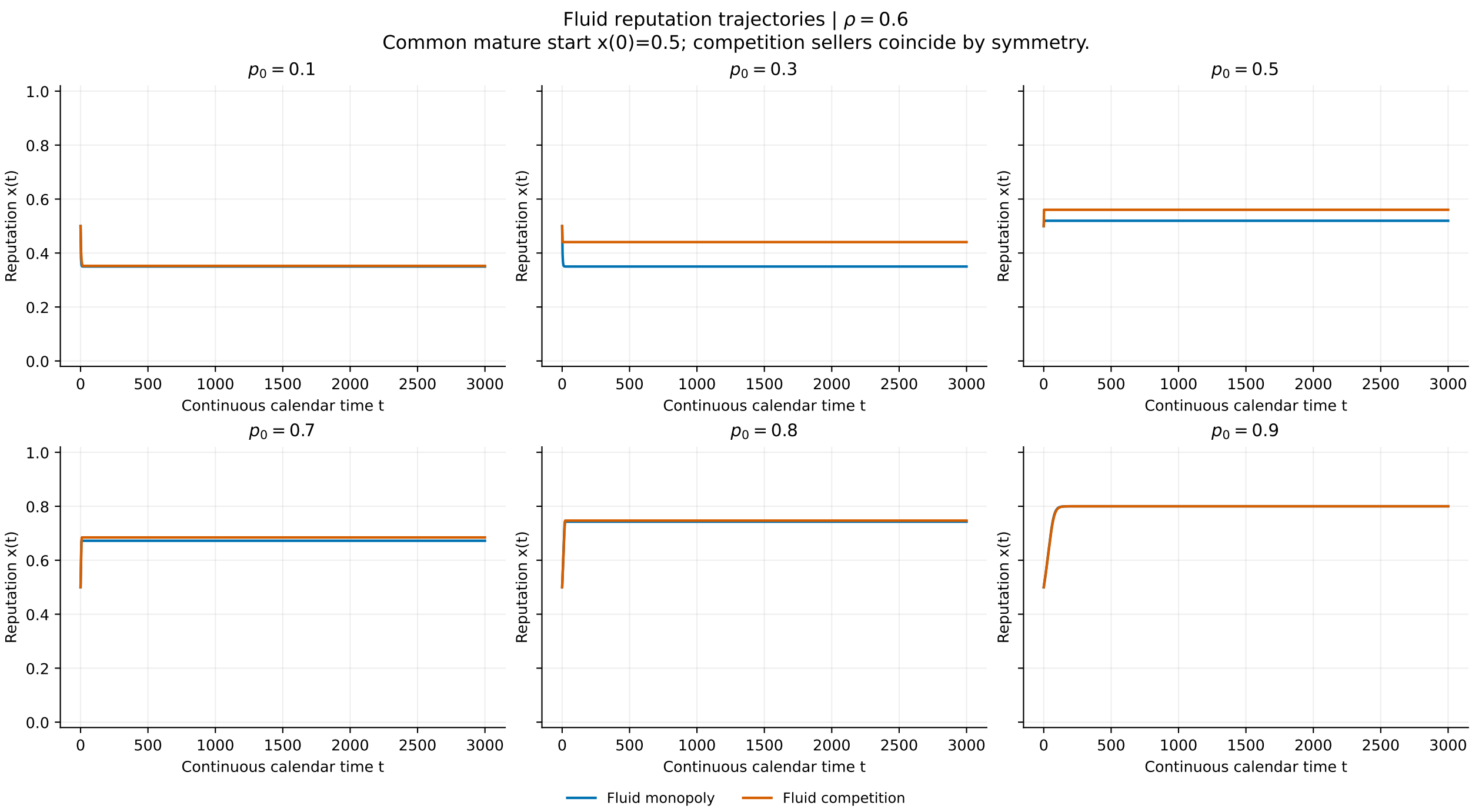

Fluid optimal policies | outside option $p_0 = 0.3$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.440565$
Competition: $\rho = 0.7$, $\bar{x} = 0.494567$
Competition: $\rho = 0.8$, $\bar{x} = 0.562469$
Competition: $\rho = 0.9$, $\bar{x} = 0.674219$

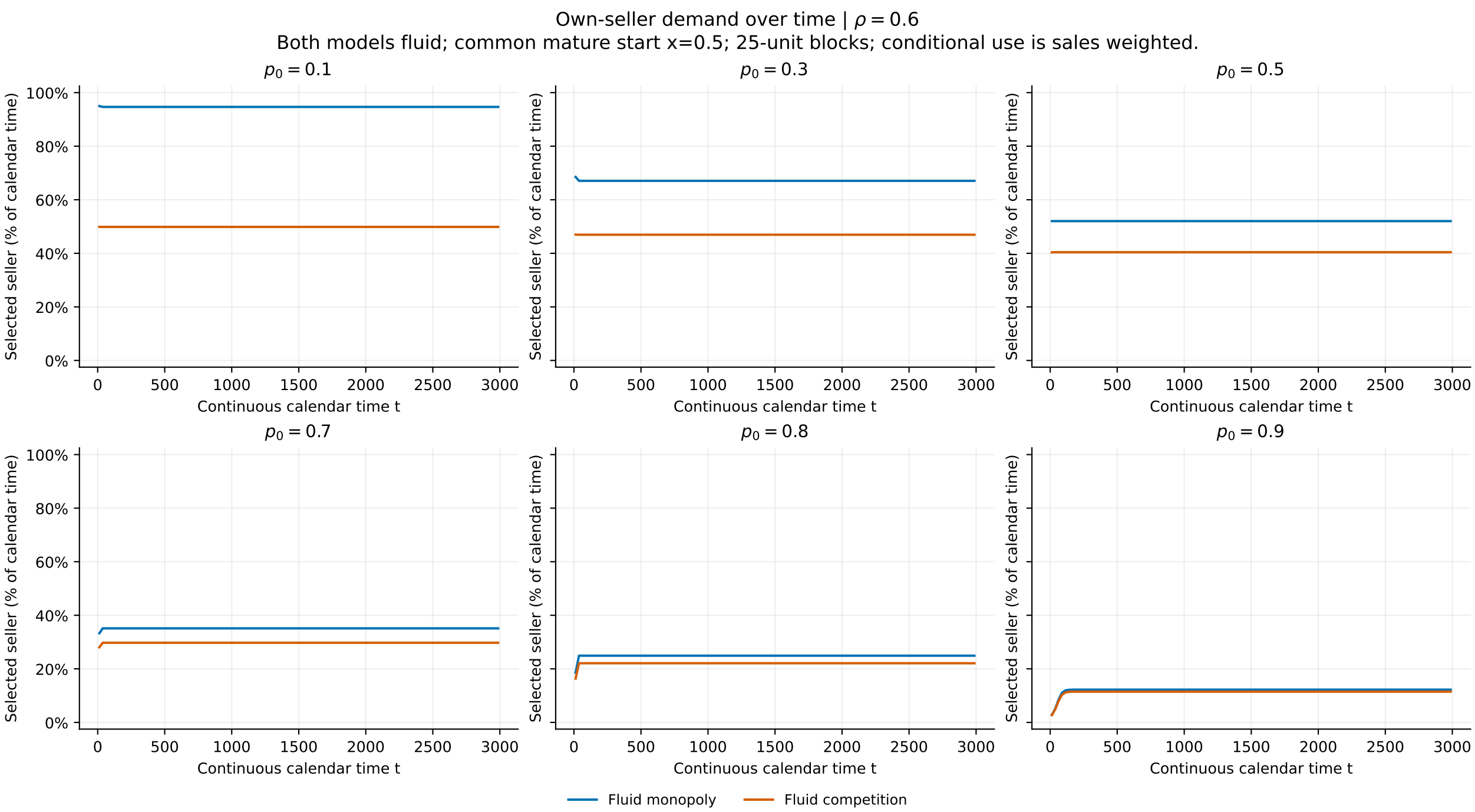

Fluid optimal policies | outside option $p_0 = 0.5$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.560326$
Competition: $\rho = 0.7$, $\bar{x} = 0.601750$
Competition: $\rho = 0.8$, $\bar{x} = 0.644722$
Competition: $\rho = 0.9$, $\bar{x} = 0.696244$

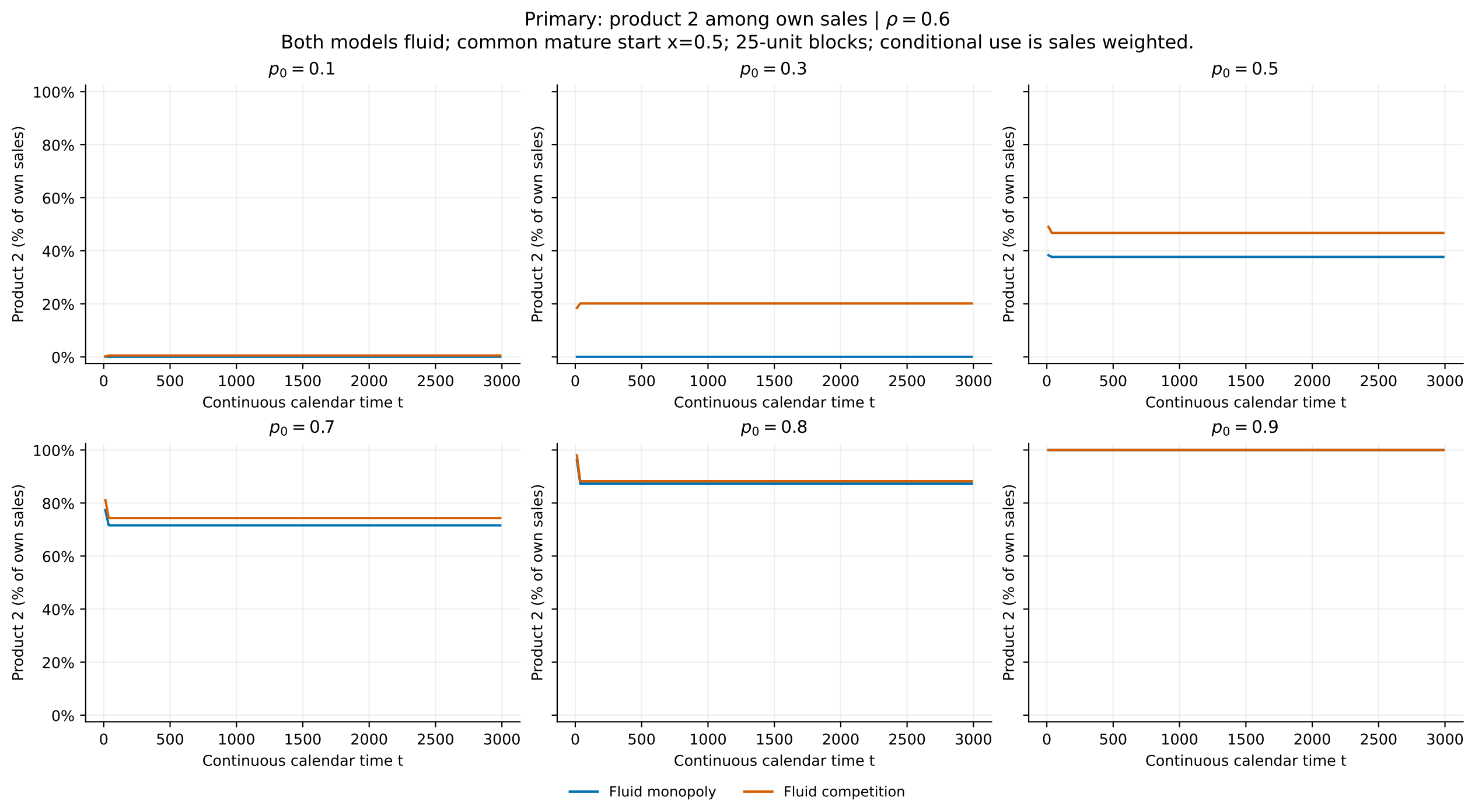

Fluid optimal policies | outside option $p_0 = 0.7$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.684454$
Competition: $\rho = 0.7$, $\bar{x} = 0.720043$
Competition: $\rho = 0.8$, $\bar{x} = 0.755493$
Competition: $\rho = 0.9$, $\bar{x} = 0.787227$

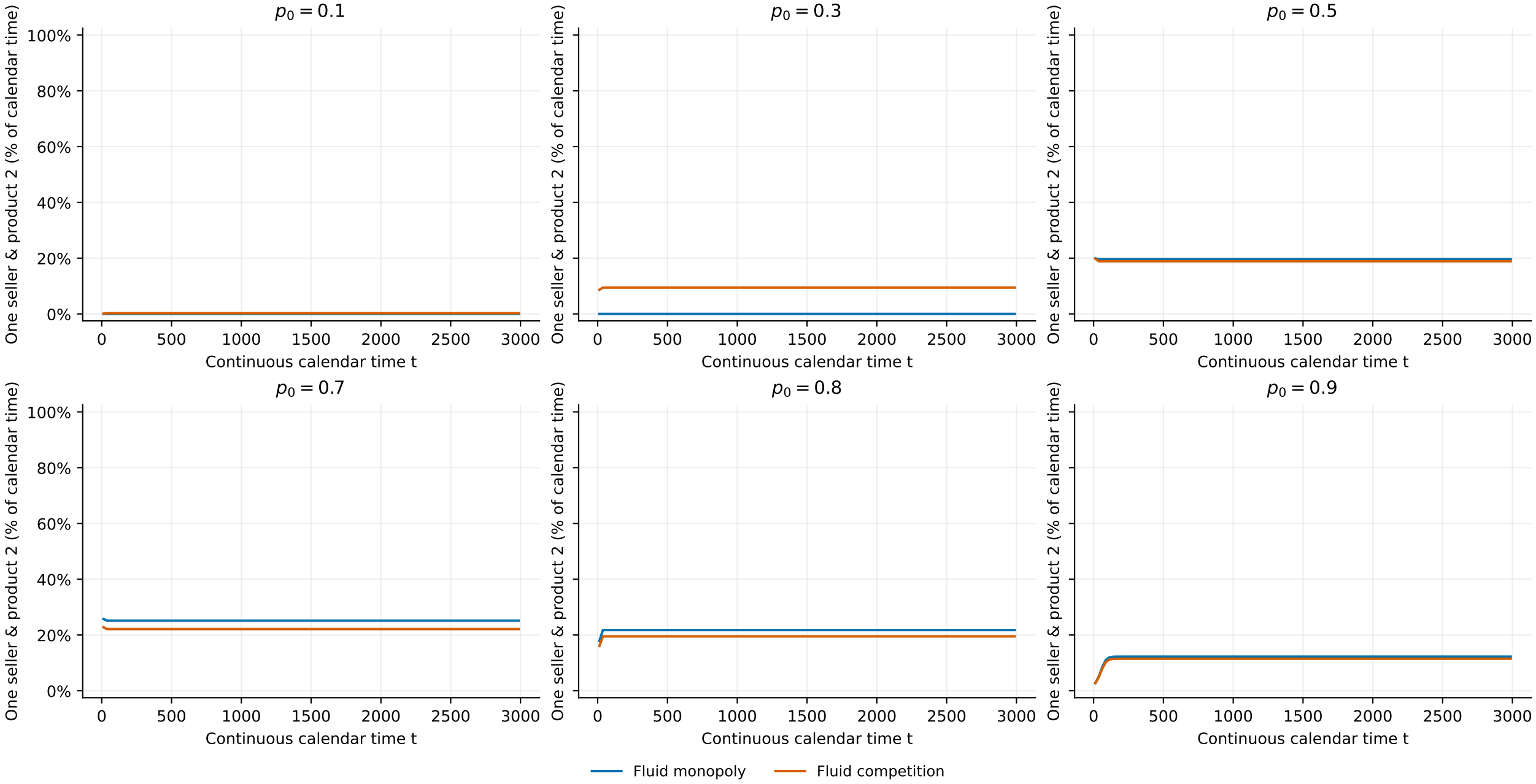

Fluid optimal policies | outside option $p_0 = 0.8$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.746980$
Competition: $\rho = 0.7$, $\bar{x} = 0.779998$
Competition: $\rho = 0.8$, $\bar{x} = 0.800000$
Competition: $\rho = 0.9$, $\bar{x} = 0.350000$
Competition: $\rho = 0.9$, $\bar{x} = 0.800000$

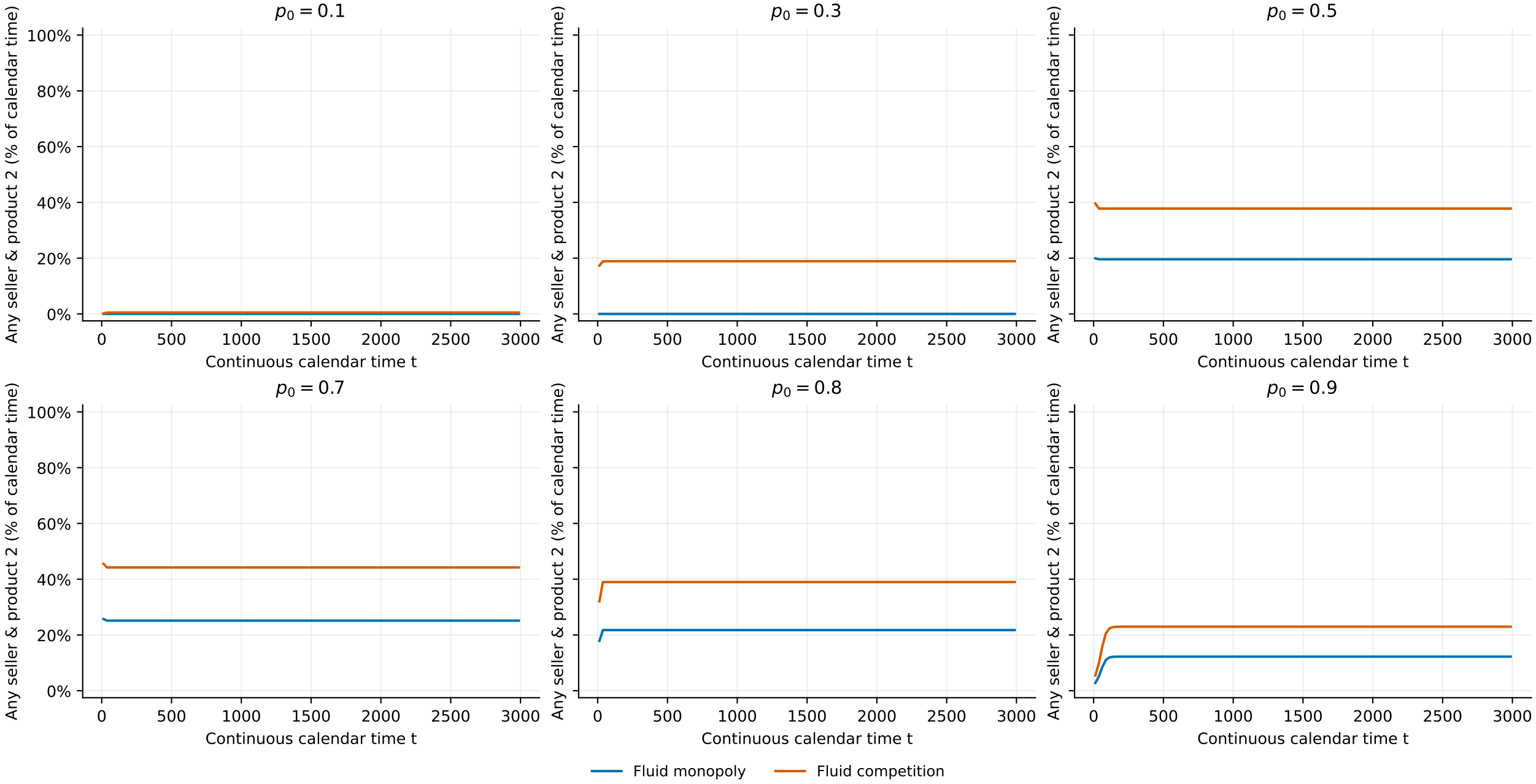

Fluid optimal policies | outside option $p_0 = 0.9$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.800000$
Competition: $\rho = 0.7$, $\bar{x} = 0.350000$
Competition: $\rho = 0.7$, $\bar{x} = 0.800000$
Competition: $\rho = 0.8$, $\bar{x} = 0.350000$
Competition: $\rho = 0.8$, $\bar{x} = 0.800000$
Competition: $\rho = 0.9$, $\bar{x} = 0.350000$

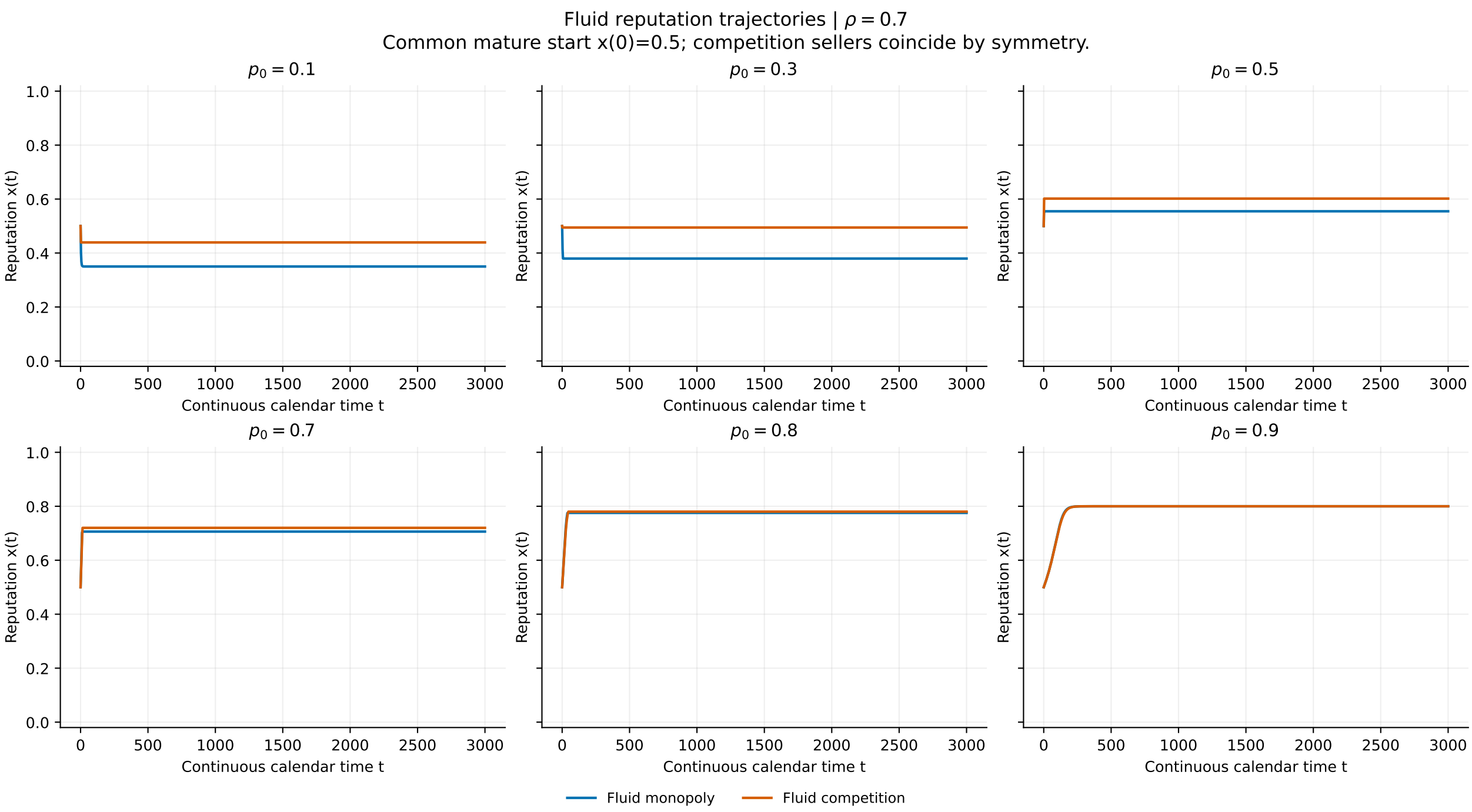

Fluid reputation trajectories | $\rho = 0.6$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

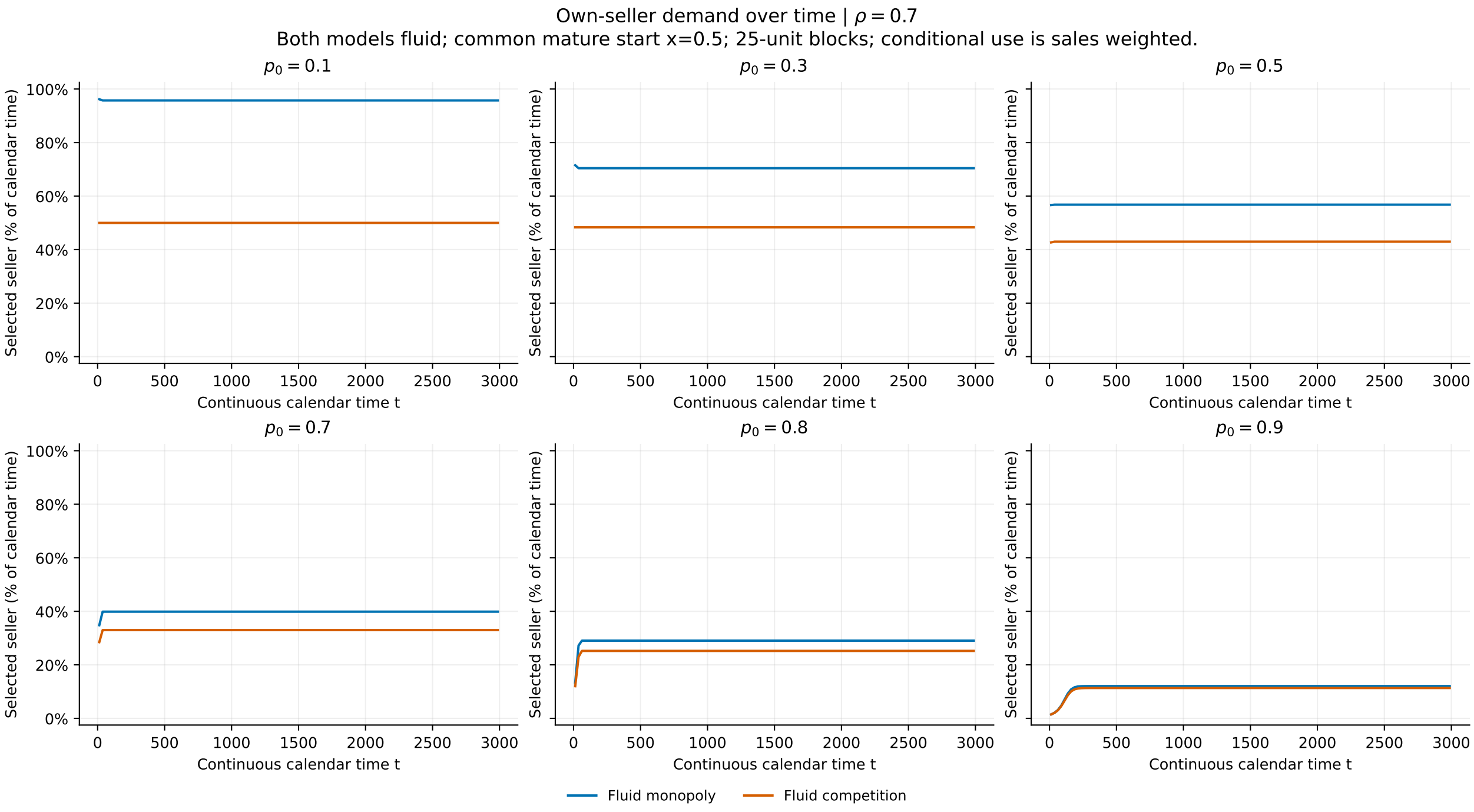

Own-seller demand over time | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

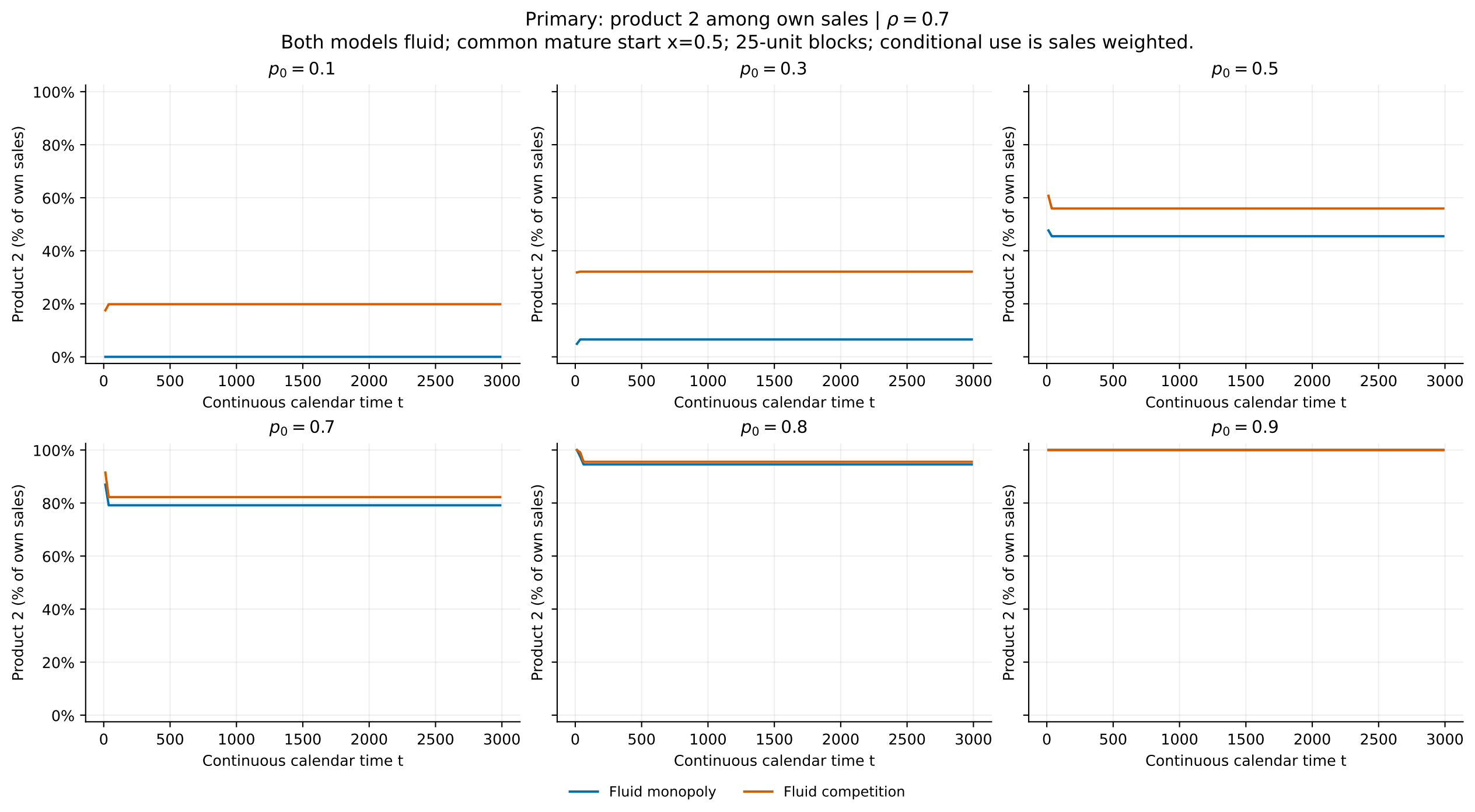

Primary: product 2 among own sales | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

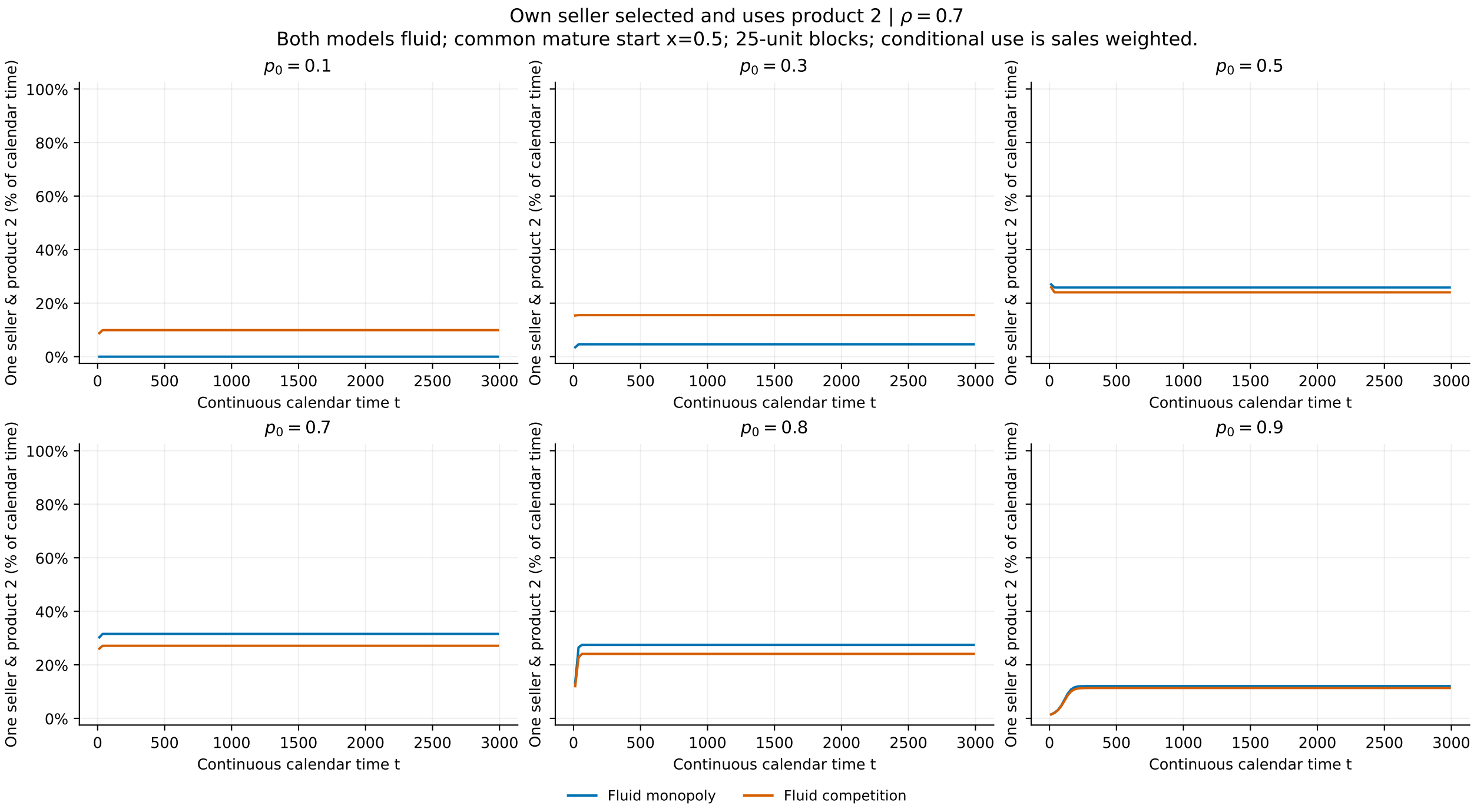

Own seller selected and uses product 2 | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

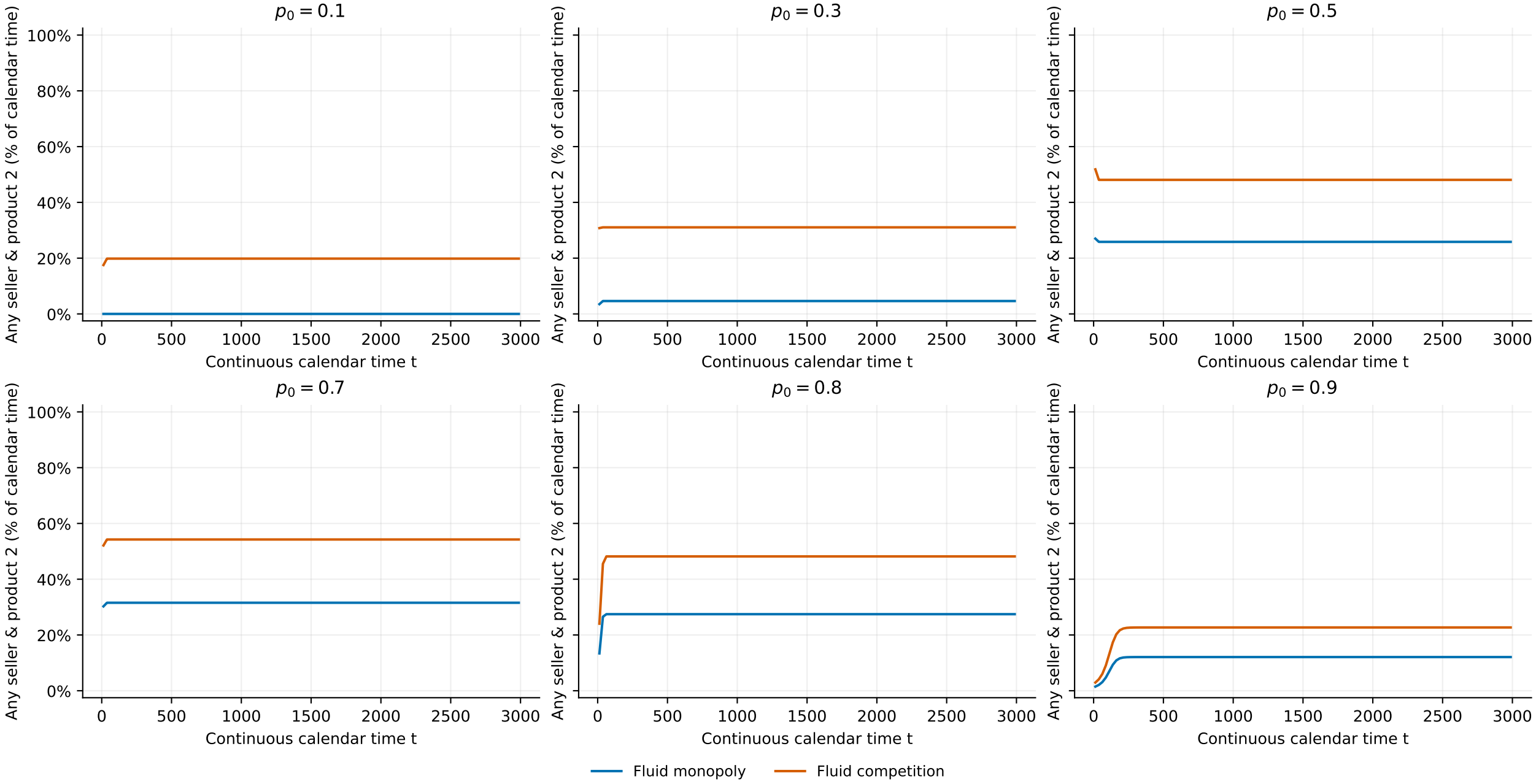

Secondary: customer receives product 2 | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

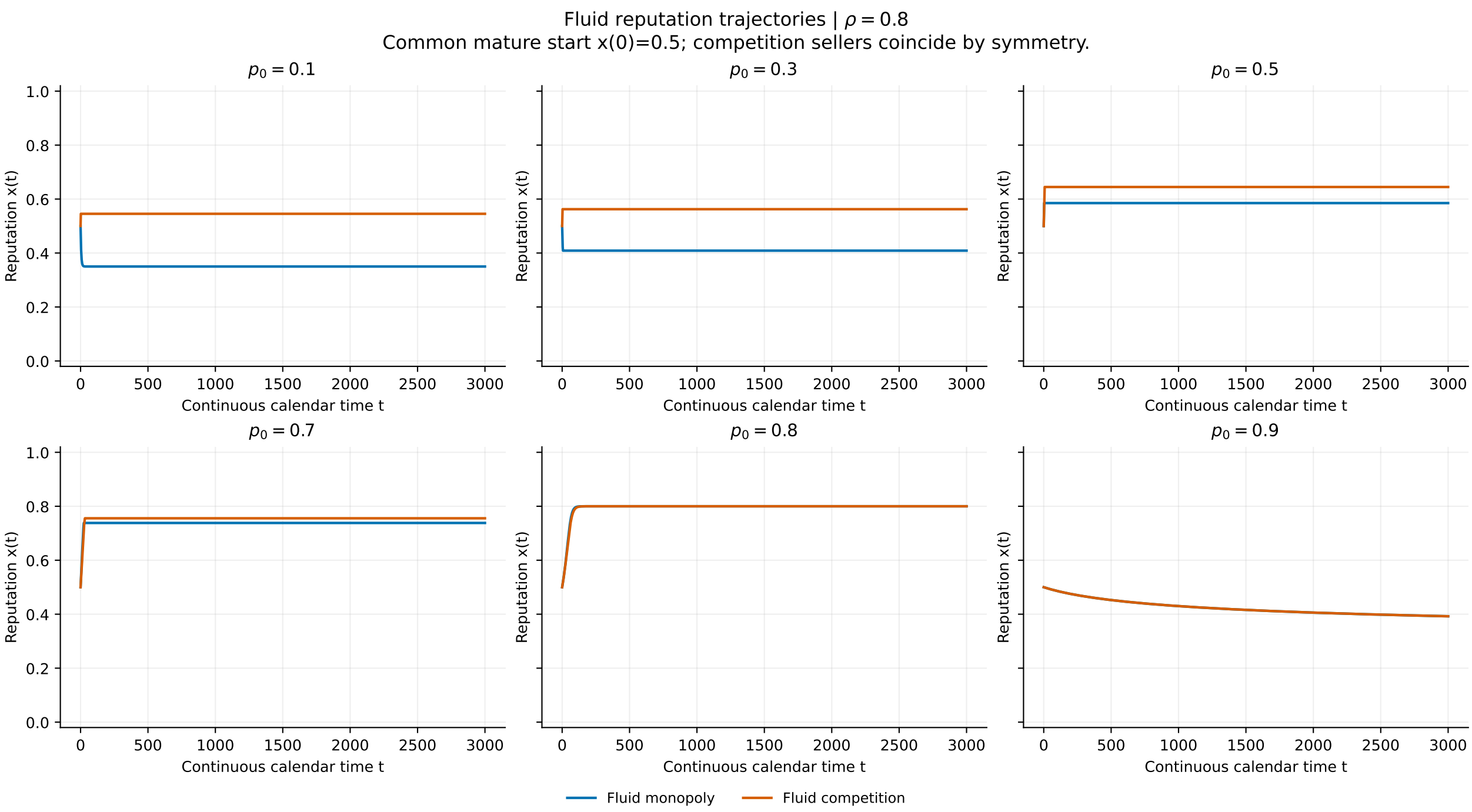

Fluid reputation trajectories | $\rho = 0.7$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

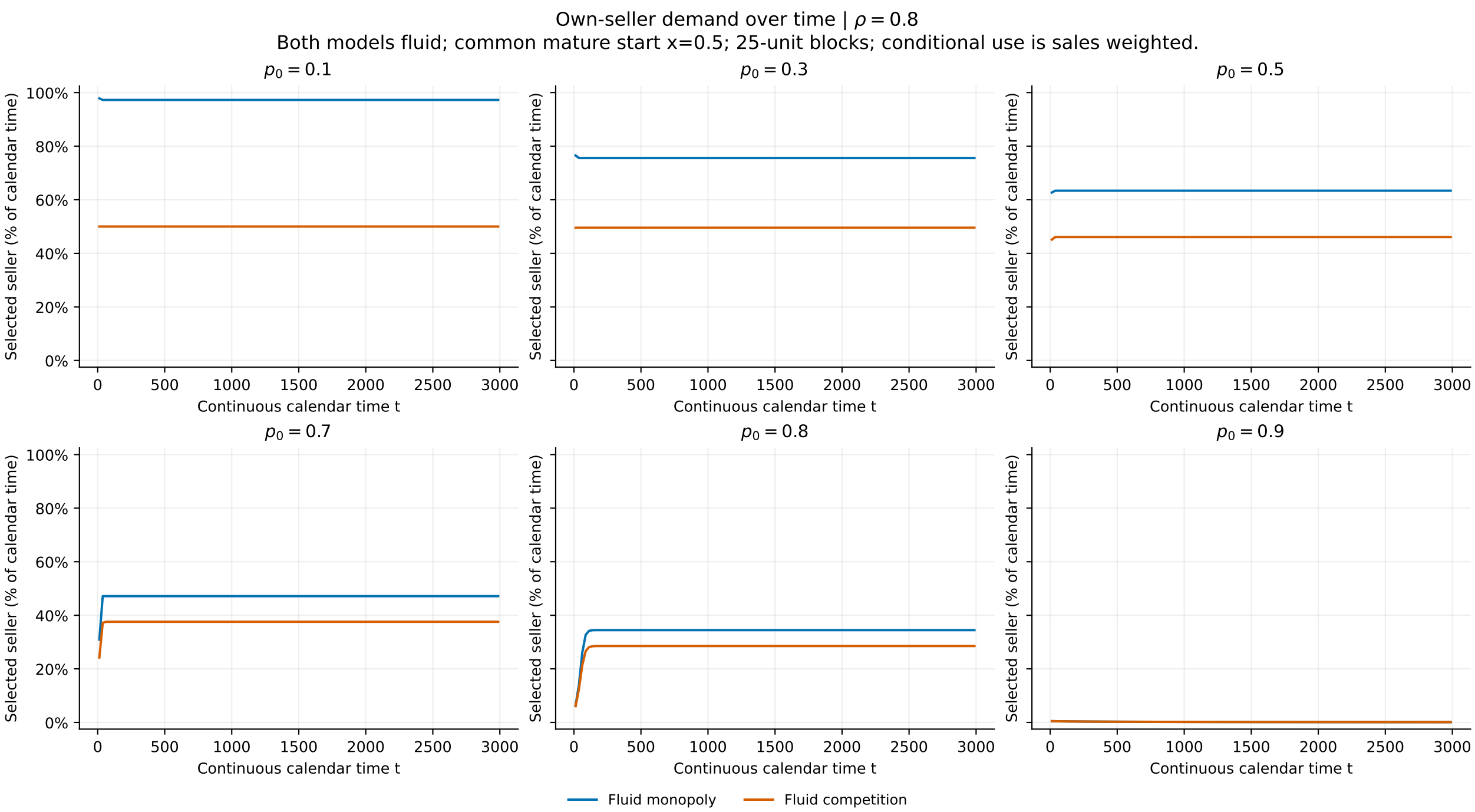

Own-seller demand over time | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

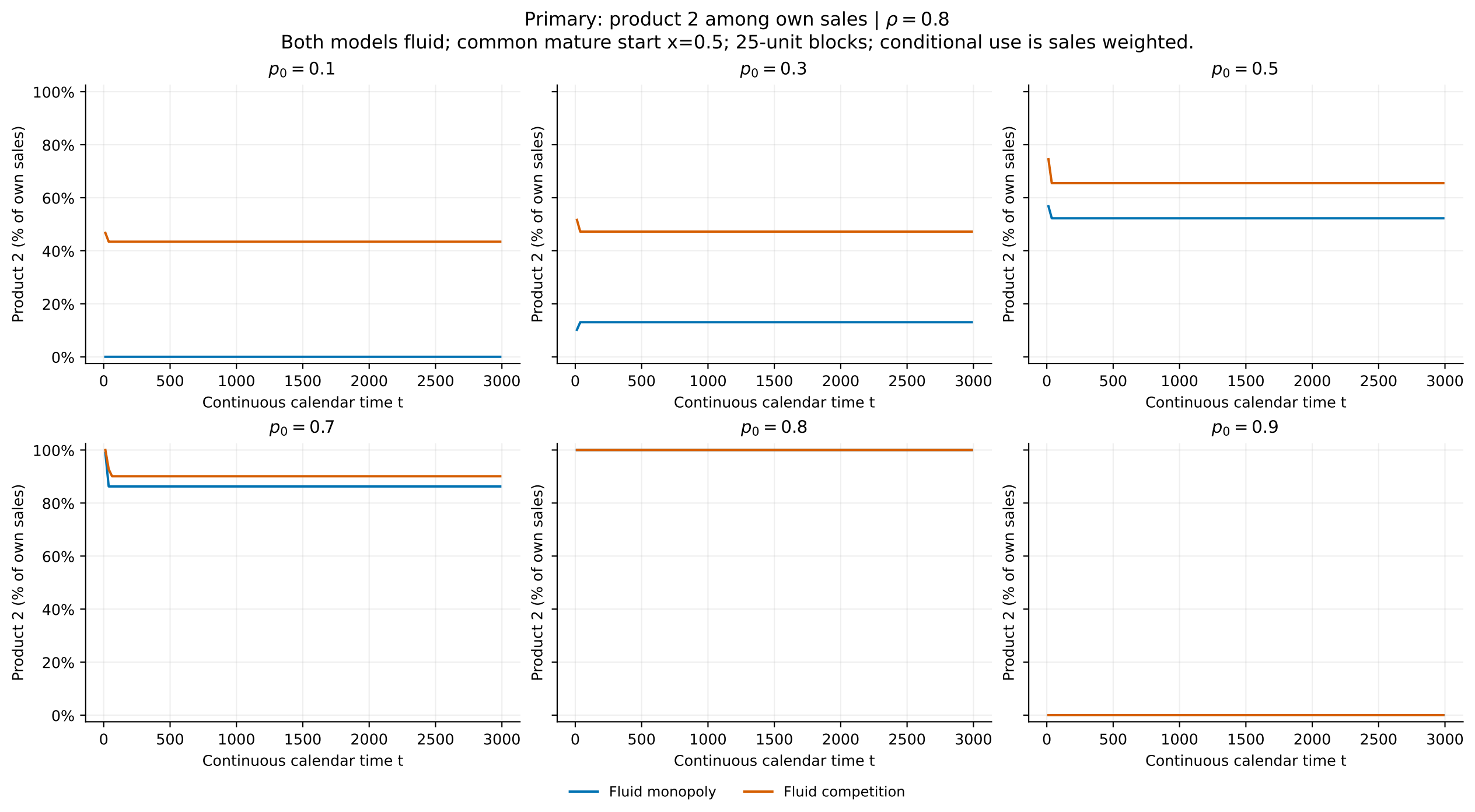

Primary: product 2 among own sales | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

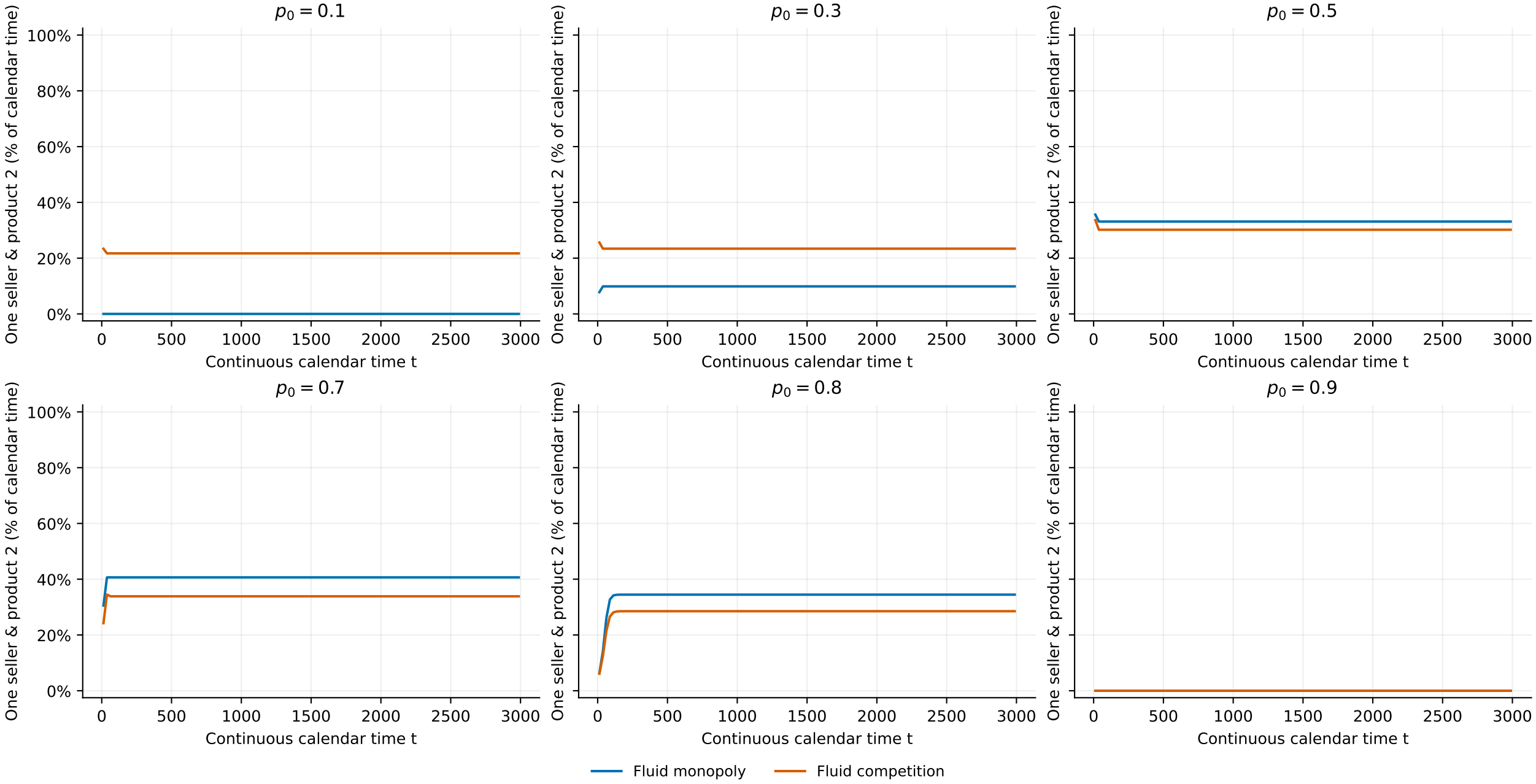

Own seller selected and uses product 2 | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
Continuous calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

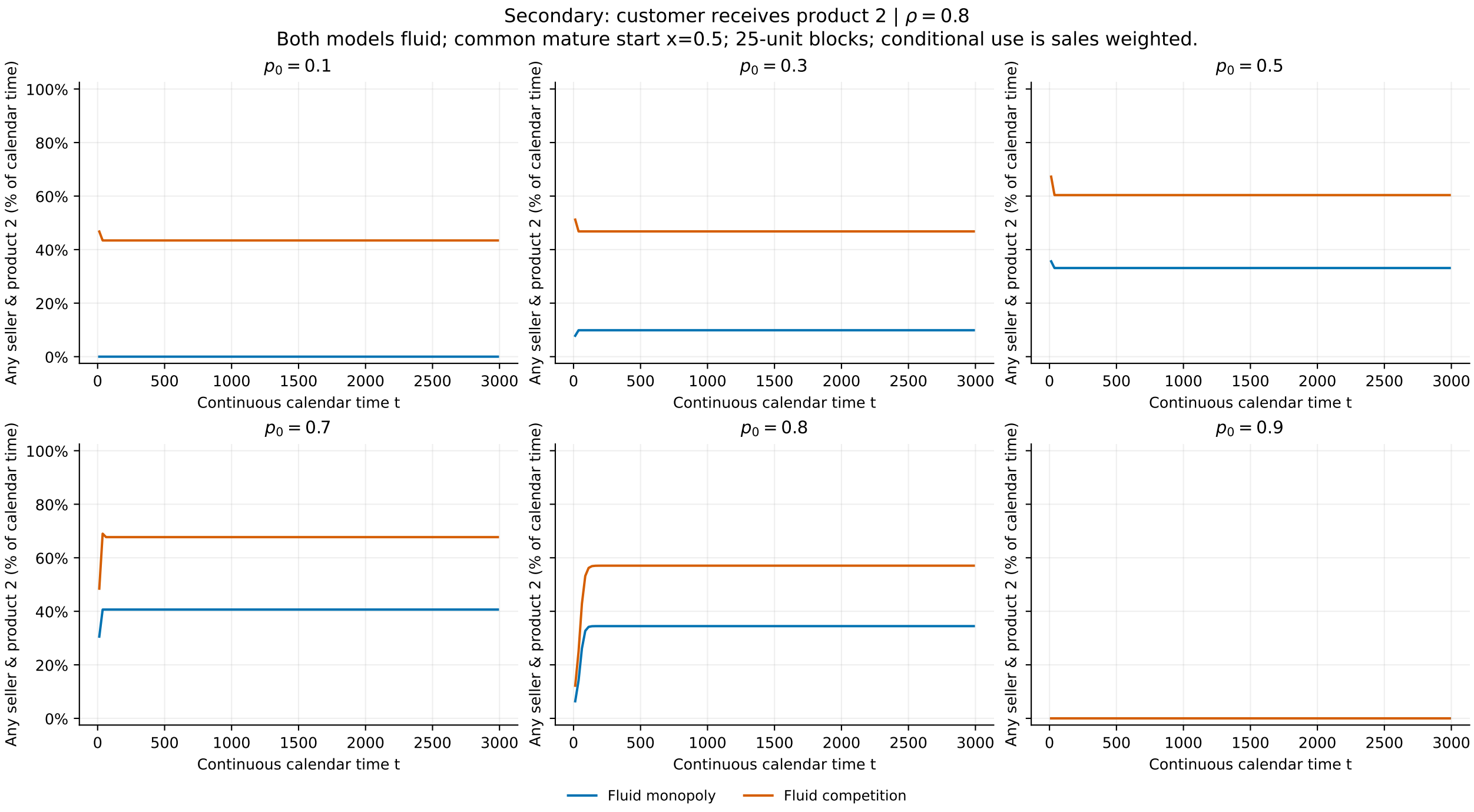

Secondary: customer receives product 2 | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

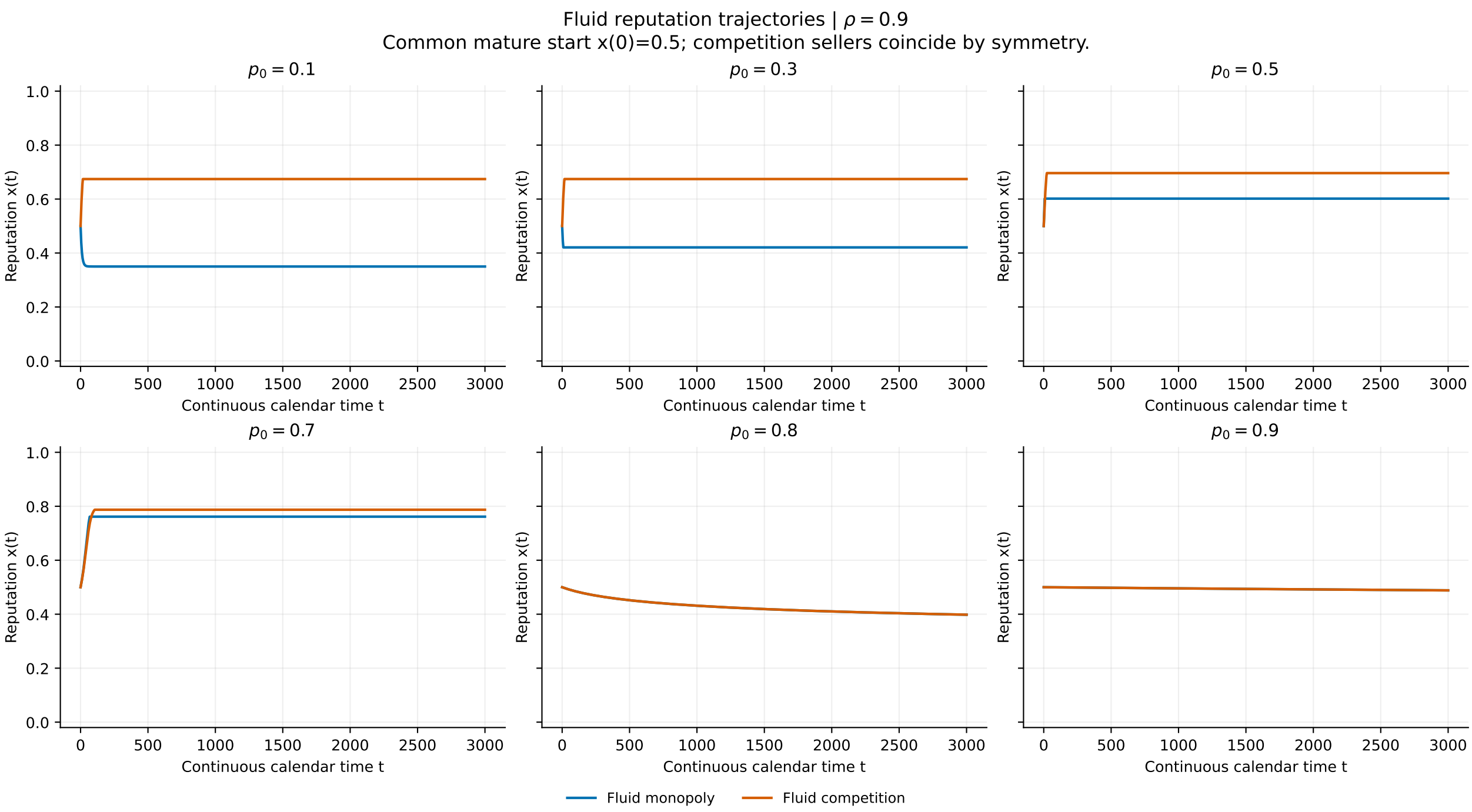

Fluid reputation trajectories | $\rho = 0.8$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

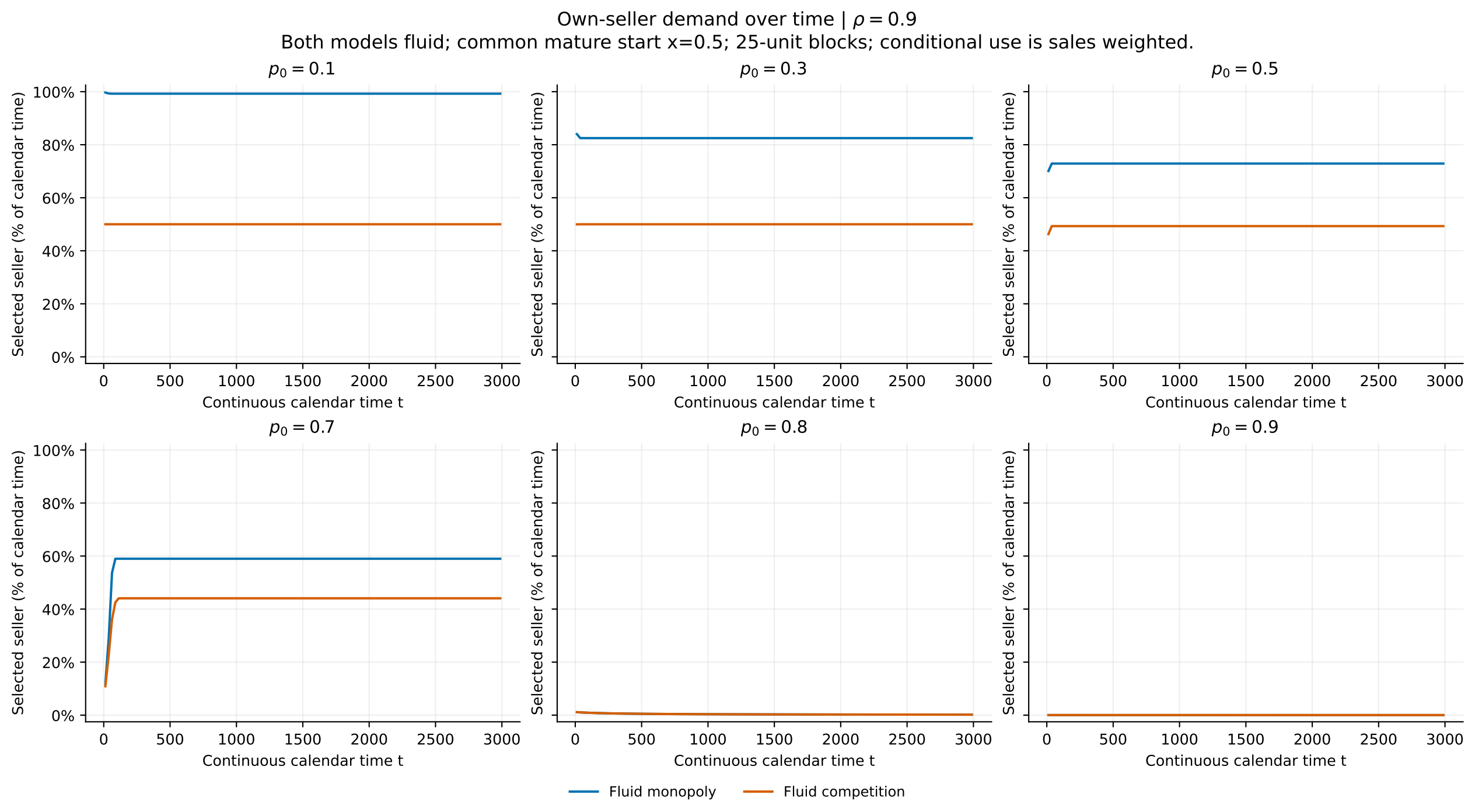

Own-seller demand over time | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

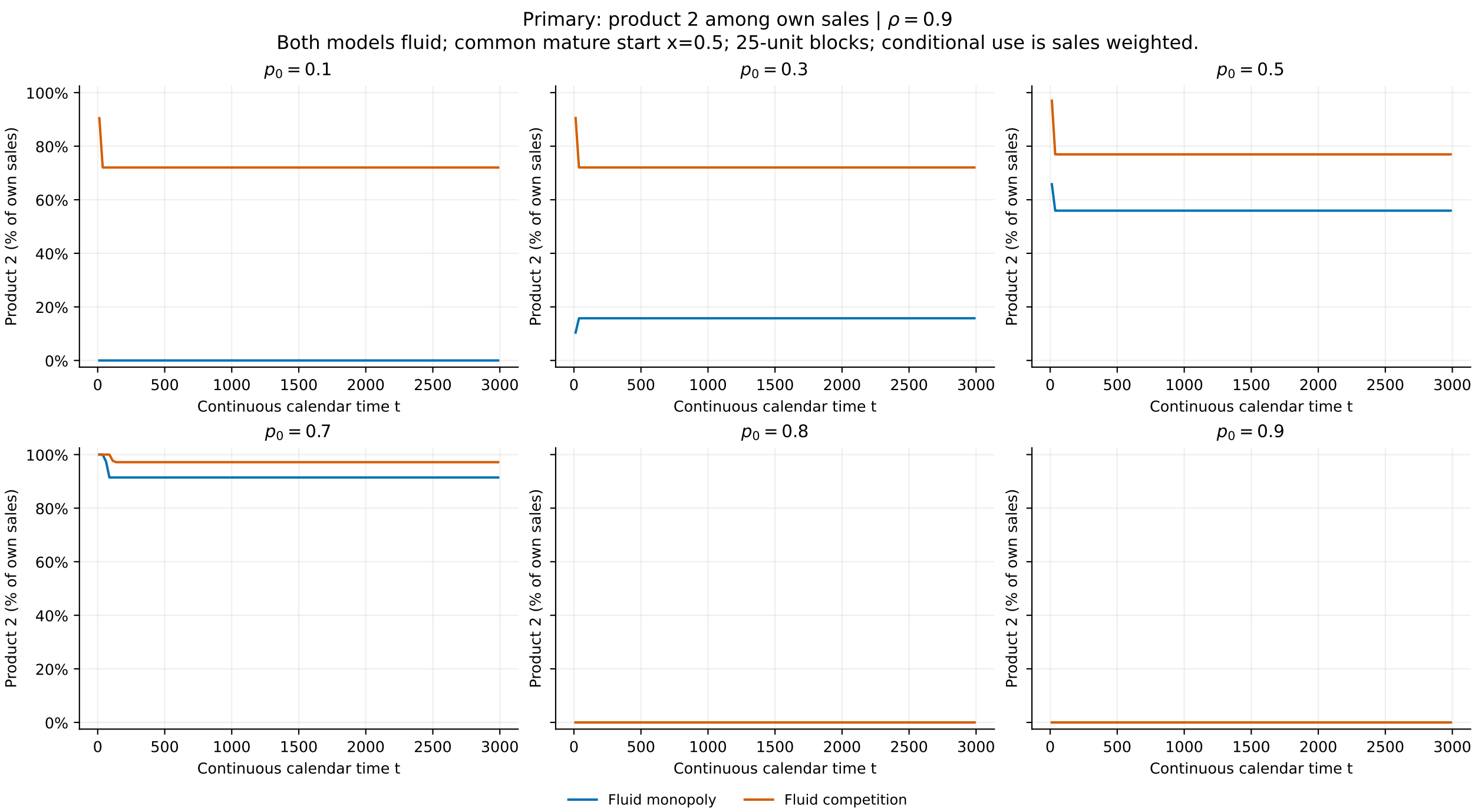

Primary: product 2 among own sales | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

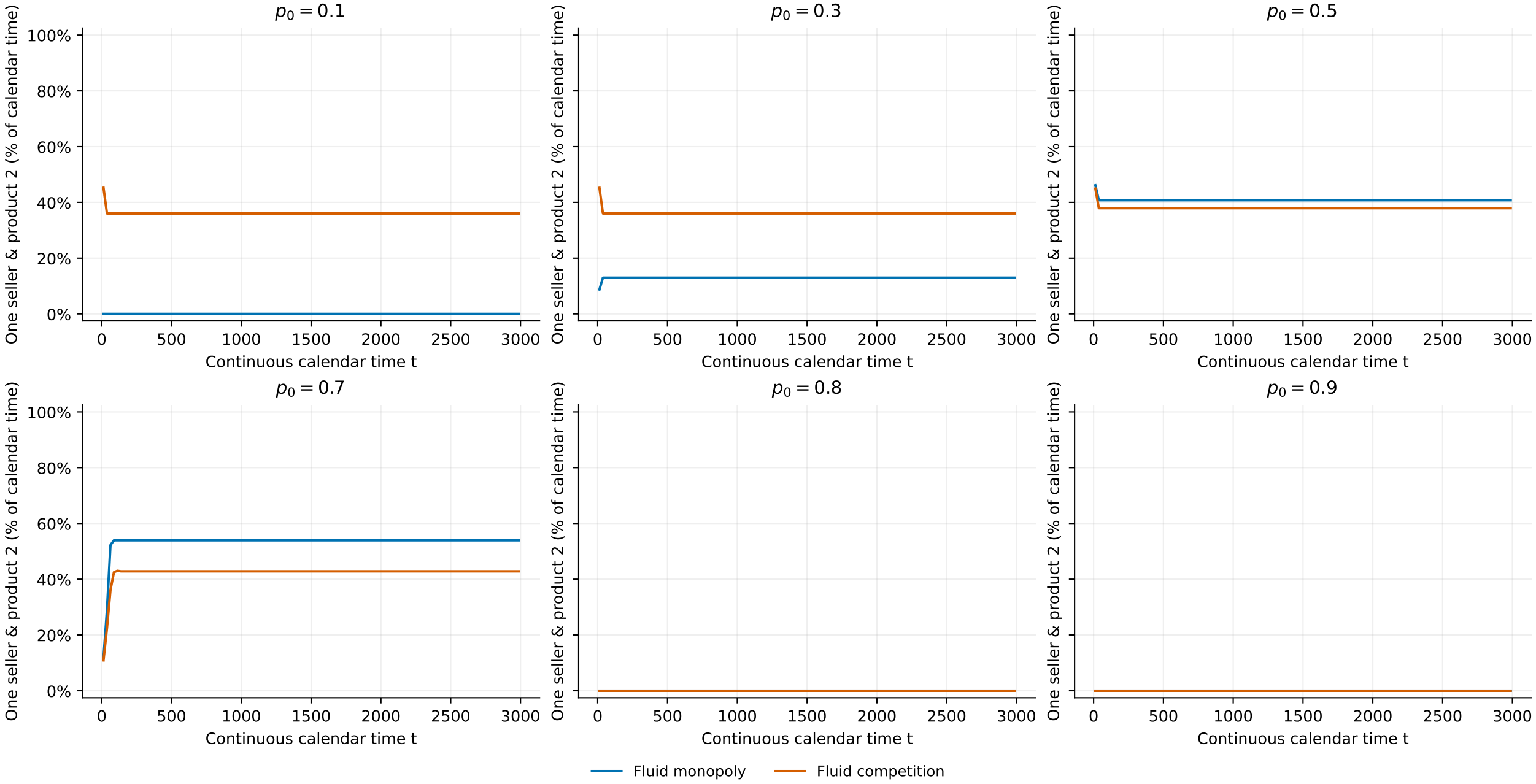

Own seller selected and uses product 2 | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

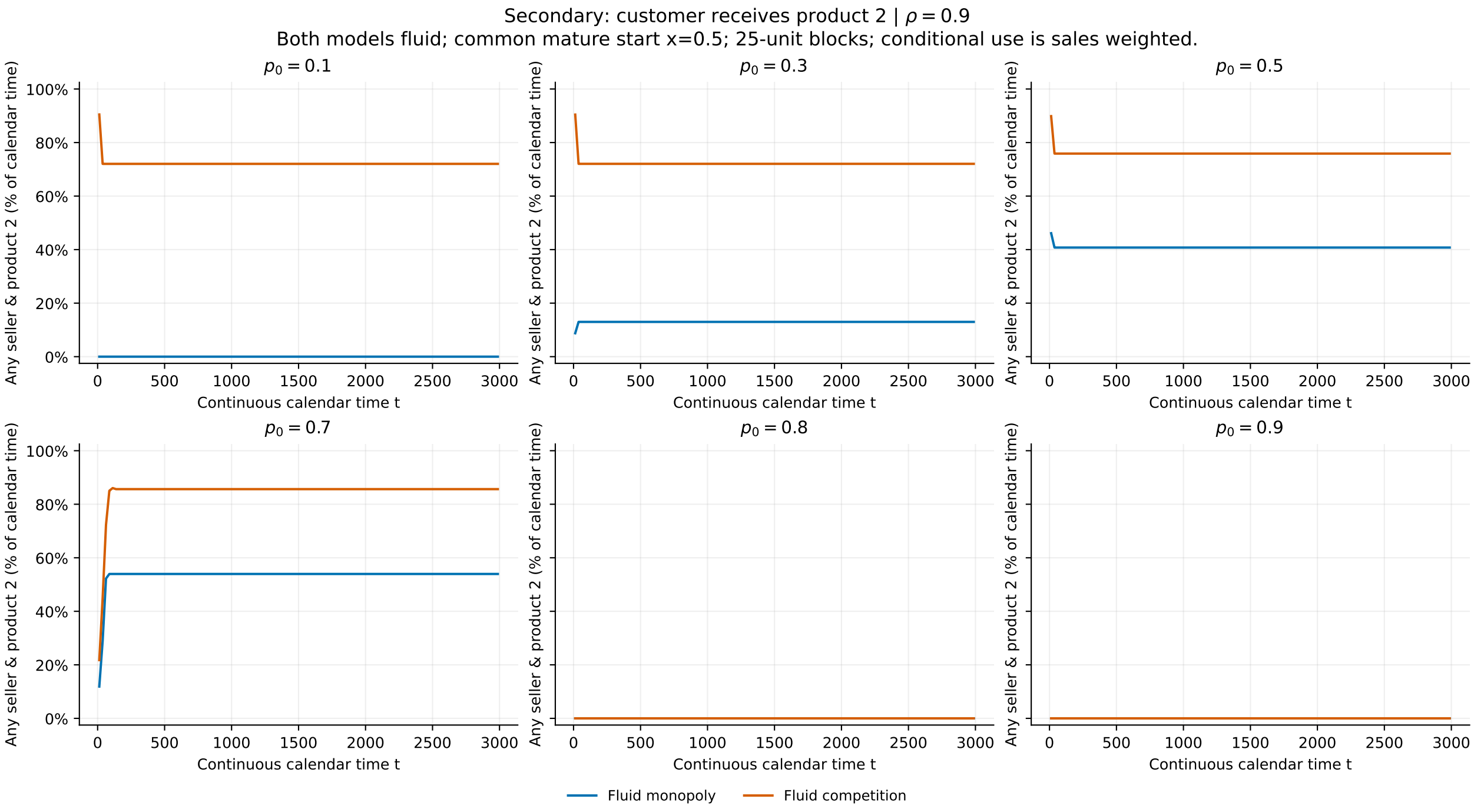

Secondary: customer receives product 2 | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

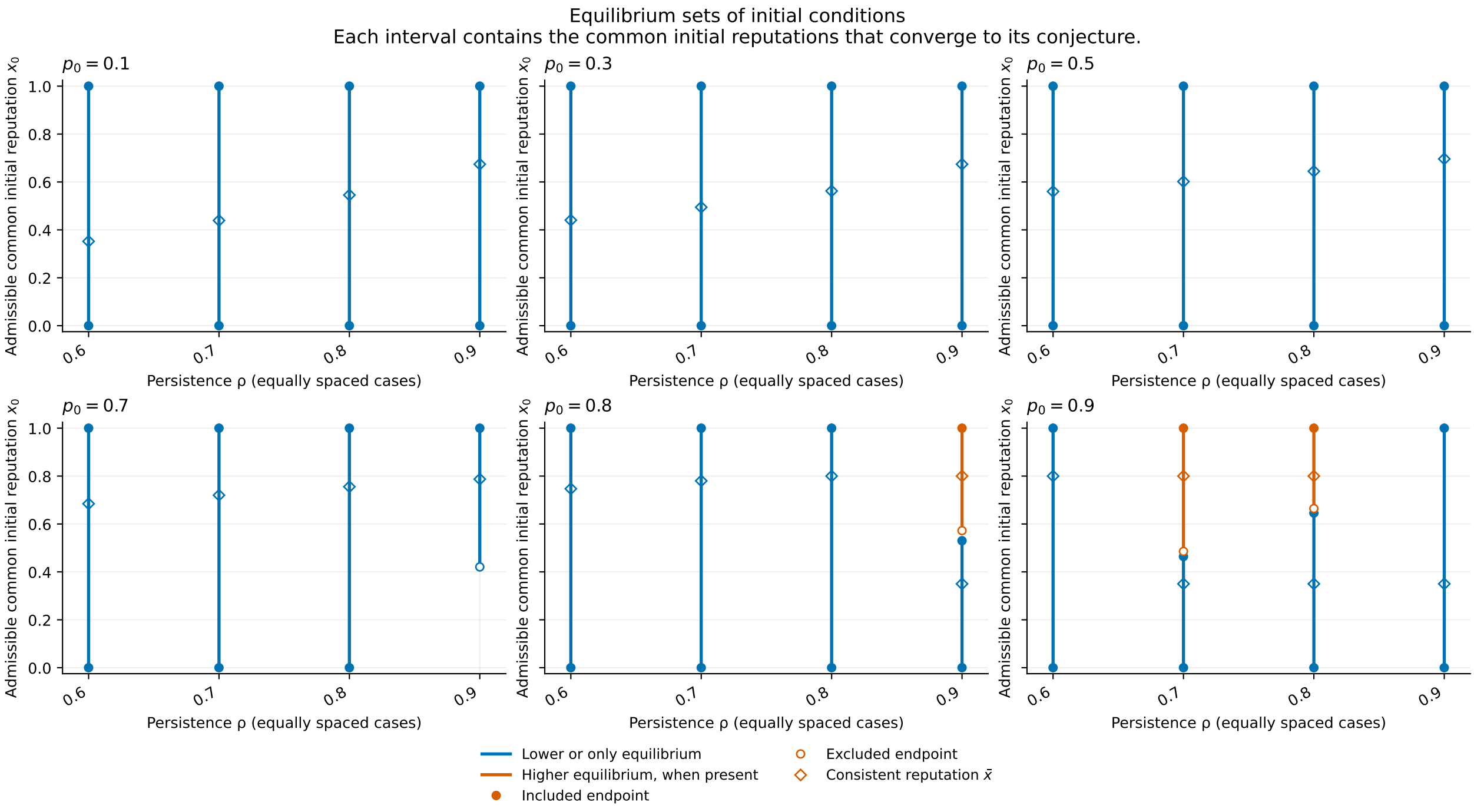

Fluid reputation trajectories | $\rho = 0.9$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

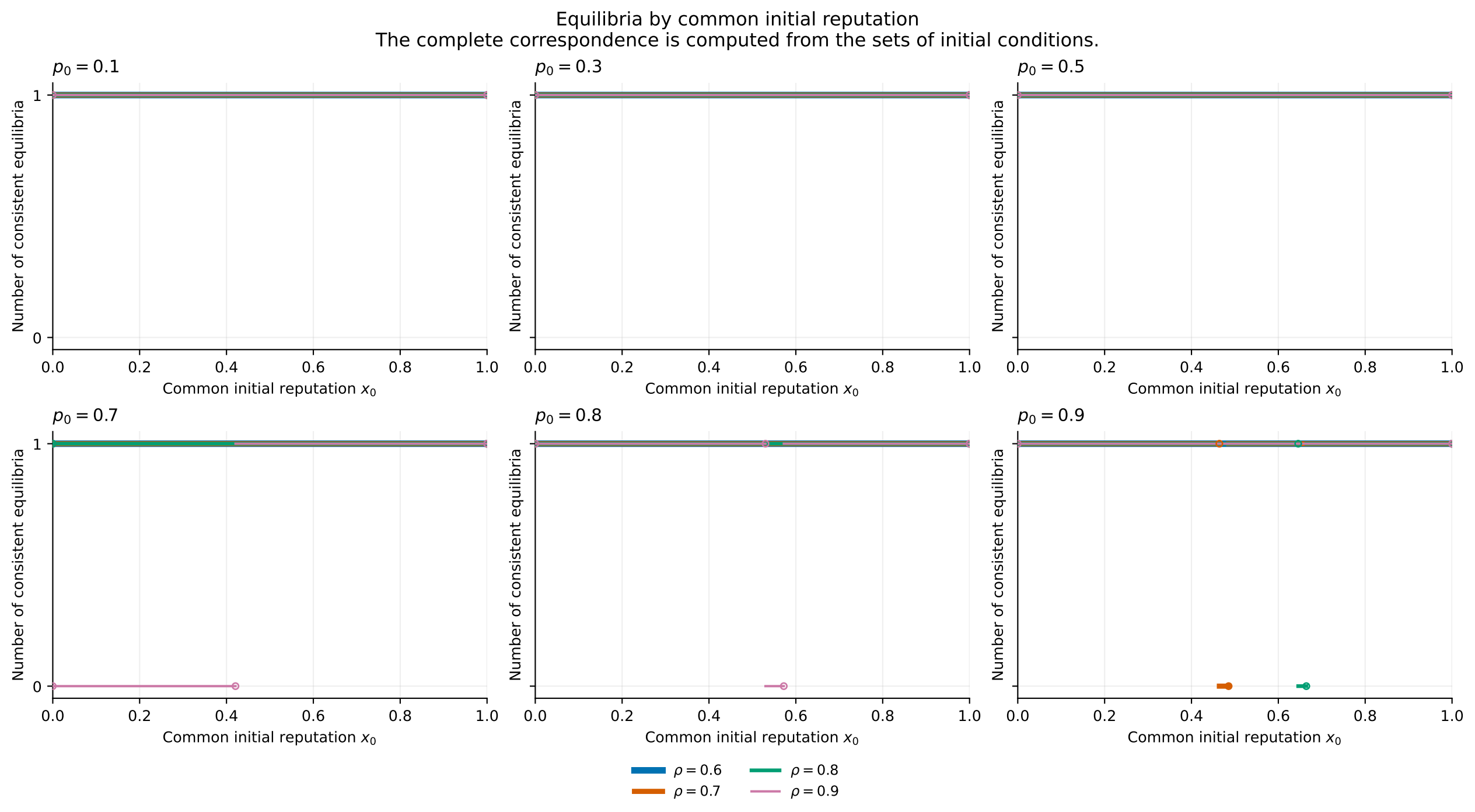

Own-seller demand over time | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

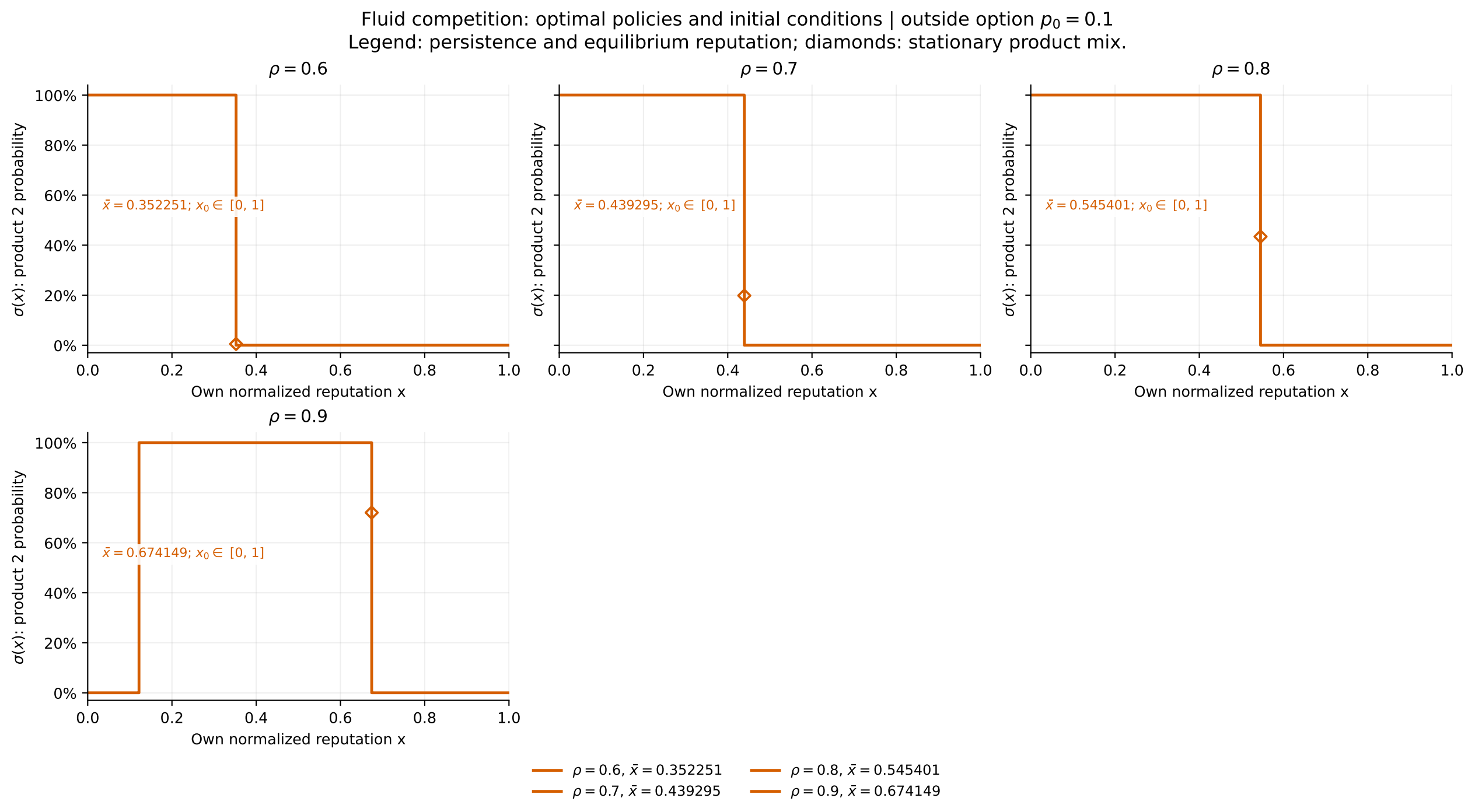

Primary: product 2 among own sales | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

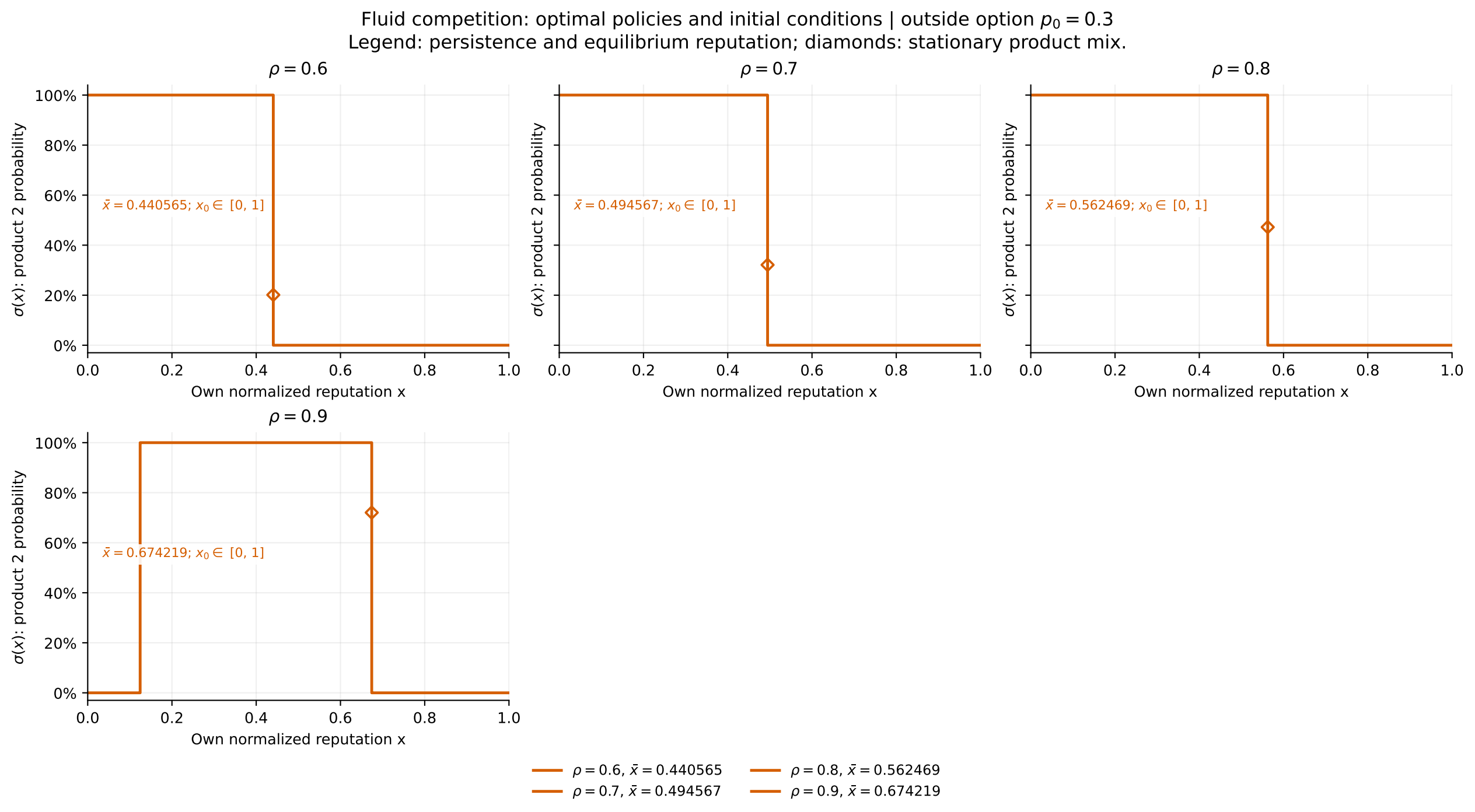

Own seller selected and uses product 2 | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

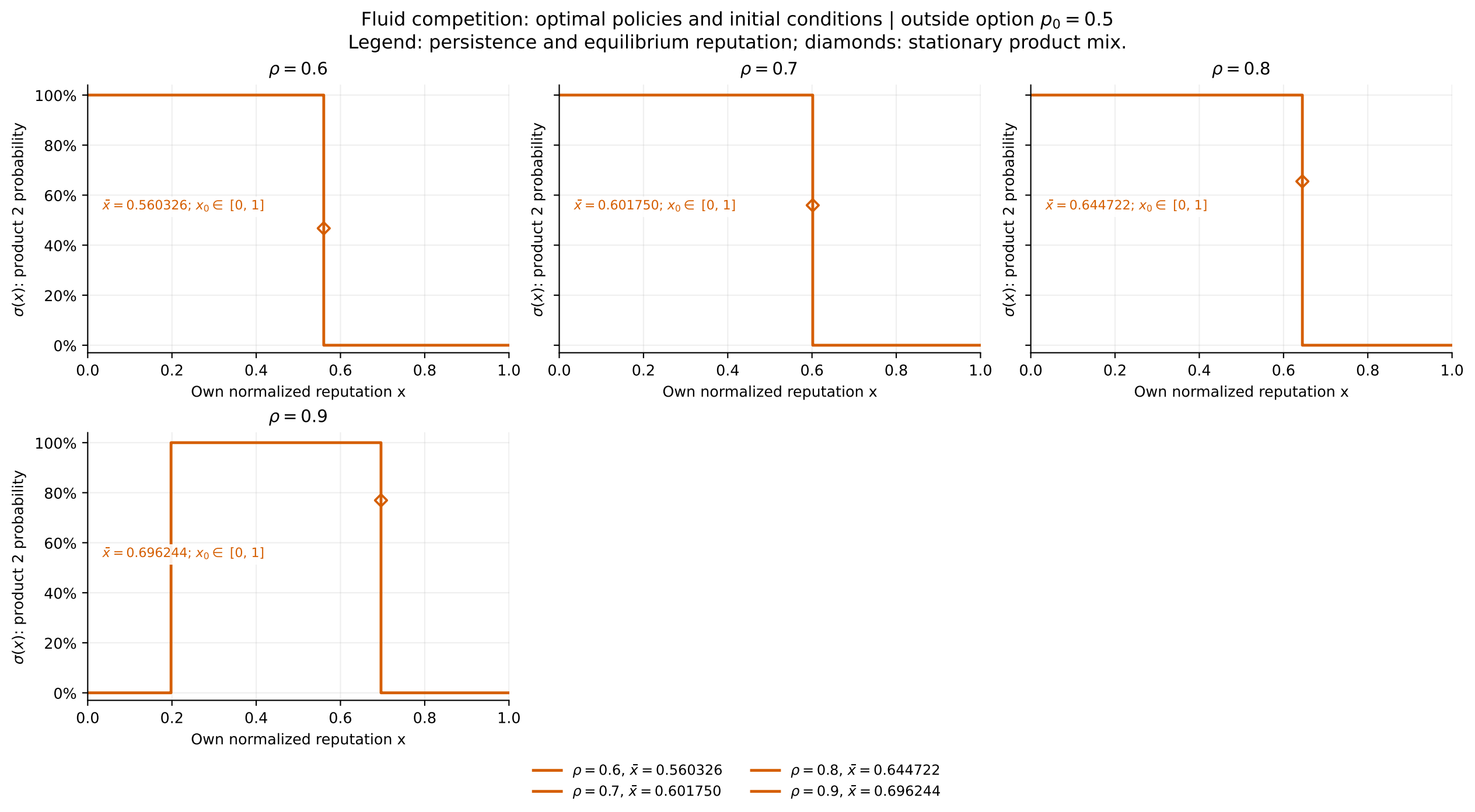

Secondary: customer receives product 2 | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

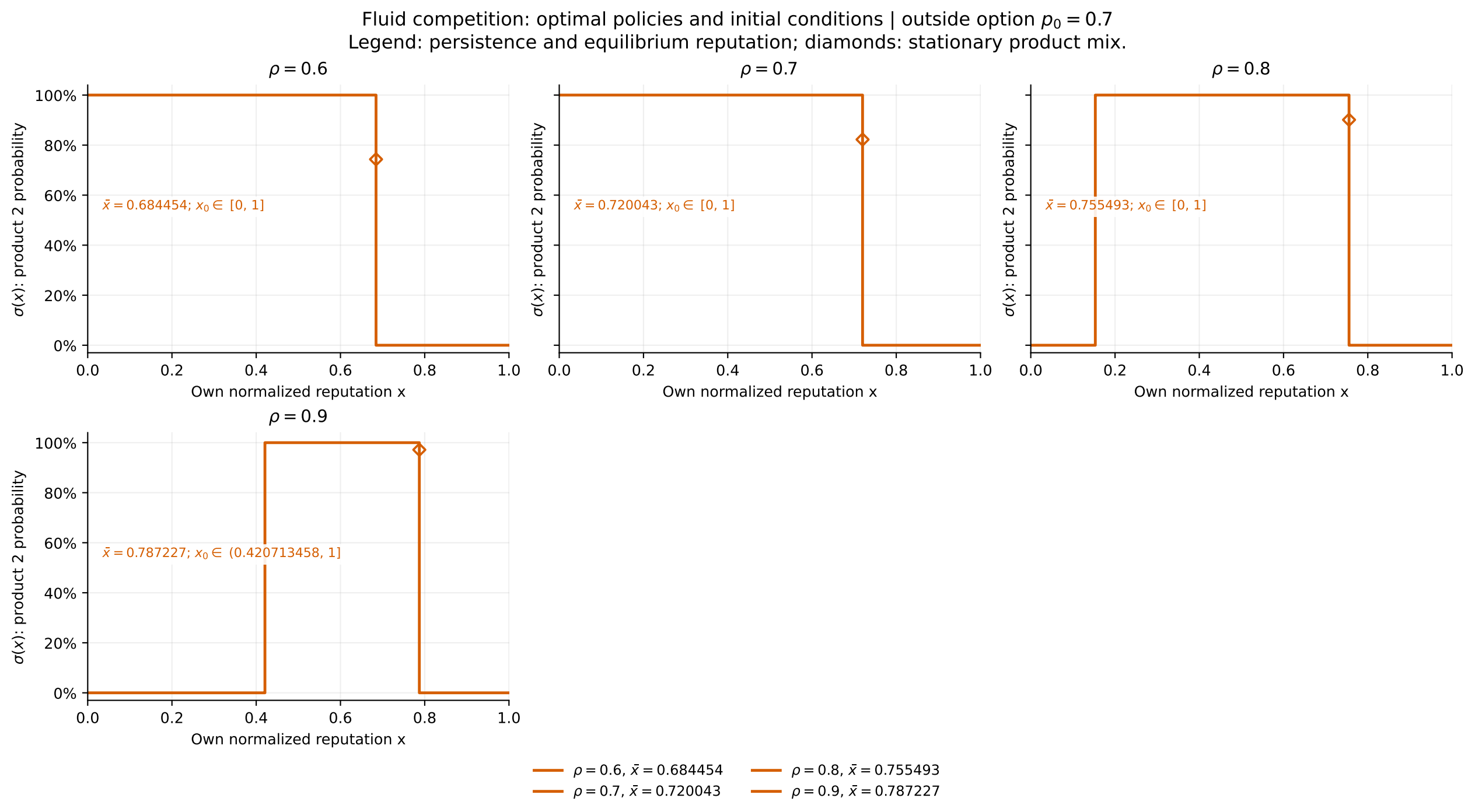

Equilibrium sets of initial conditions
Each interval contains the common initial reputations that converge to its conjecture.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Admissible common initial reputation x_0
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Lower or only equilibrium
Higher equilibrium, when present
Included endpoint
Excluded endpoint
Consistent reputation $\bar{x}$

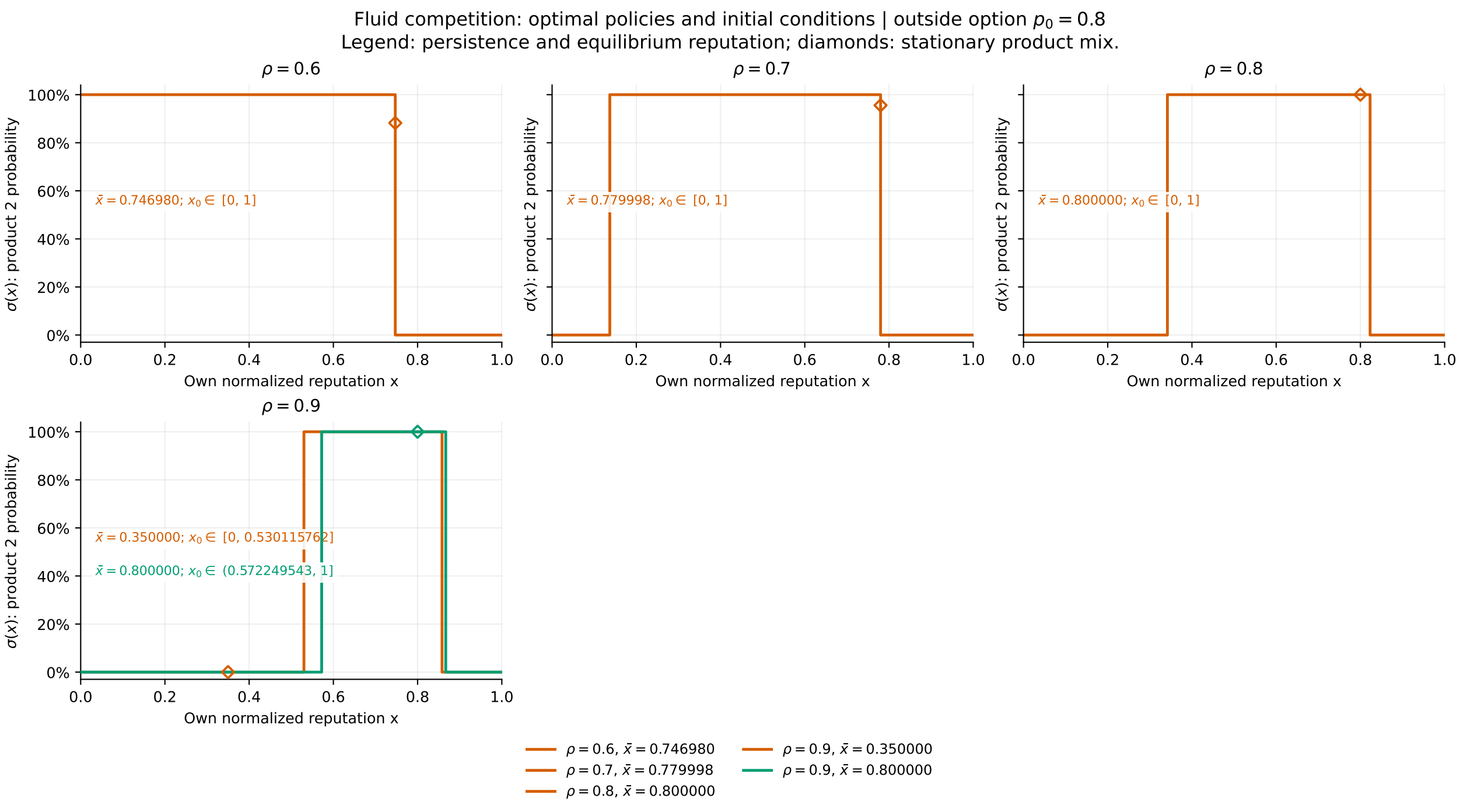

Equilibria by common initial reputation
The complete correspondence is computed from the sets of initial conditions.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Number of consistent equilibria
Common initial reputation x_0
0
1
0.0
0.2
0.4
0.6
0.8
1.0
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$

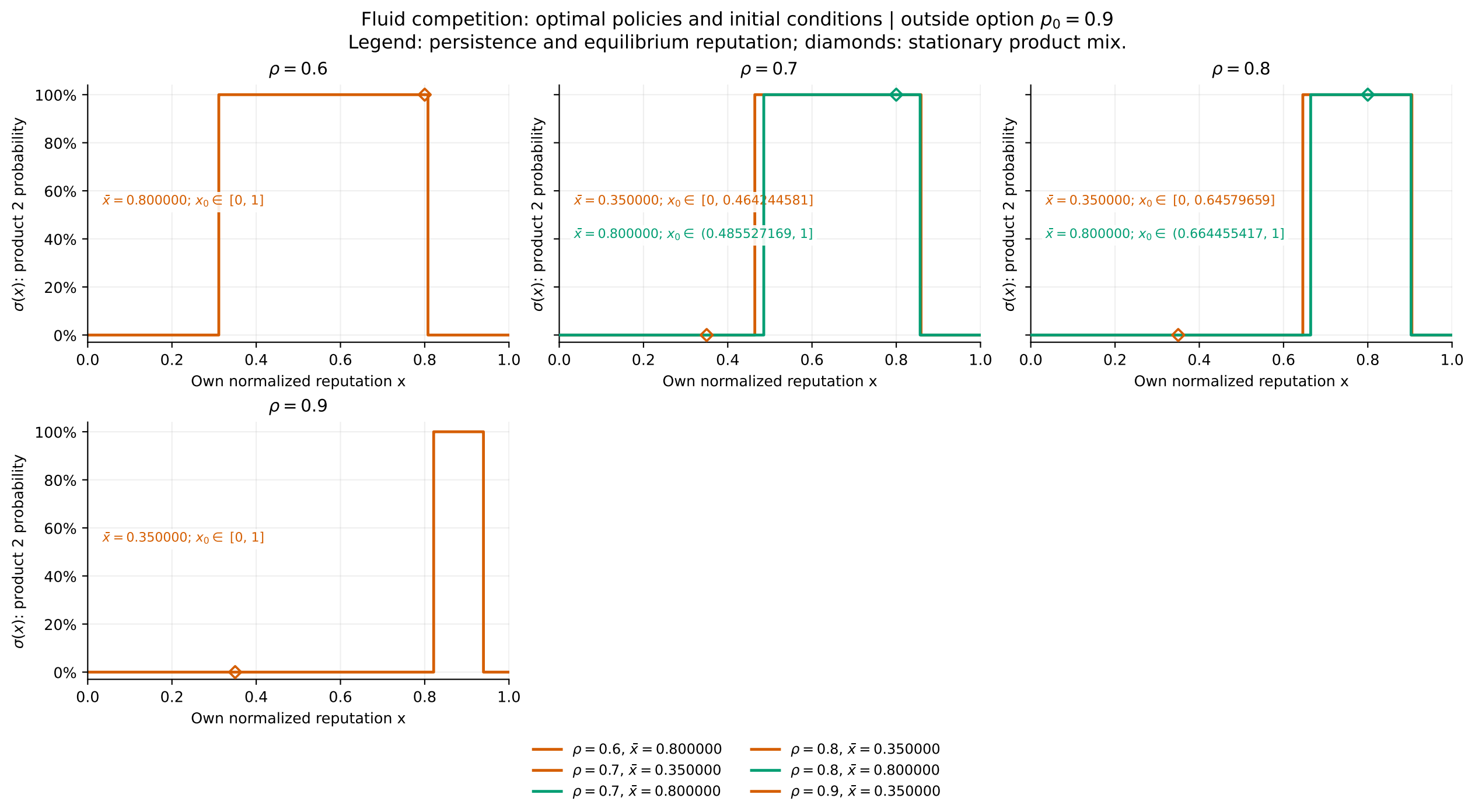

Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.1$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.352251$; $x_0 \in [0, 1]$
$\bar{x} = 0.439295$; $x_0 \in [0, 1]$
$\bar{x} = 0.545401$; $x_0 \in [0, 1]$
$\bar{x} = 0.674149$; $x_0 \in [0, 1]$
$\rho = 0.6$, $\bar{x} = 0.352251$
$\rho = 0.7$, $\bar{x} = 0.439295$
$\rho = 0.8$, $\bar{x} = 0.545401$
$\rho = 0.9$, $\bar{x} = 0.674149$


Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.3$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.440565$; $x_0 \in [0, 1]$
$\bar{x} = 0.494567$; $x_0 \in [0, 1]$
$\bar{x} = 0.562469$; $x_0 \in [0, 1]$
$\bar{x} = 0.674219$; $x_0 \in [0, 1]$
$\rho = 0.6$, $\bar{x} = 0.440565$
$\rho = 0.7$, $\bar{x} = 0.494567$
$\rho = 0.8$, $\bar{x} = 0.562469$
$\rho = 0.9$, $\bar{x} = 0.674219$


Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.5$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.560326$; $x_0 \in [0, 1]$
$\bar{x} = 0.601750$; $x_0 \in [0, 1]$
$\bar{x} = 0.644722$; $x_0 \in [0, 1]$
$\bar{x} = 0.696244$; $x_0 \in [0, 1]$
$\rho = 0.6$, $\bar{x} = 0.560326$
$\rho = 0.7$, $\bar{x} = 0.601750$
$\rho = 0.8$, $\bar{x} = 0.644722$
$\rho = 0.9$, $\bar{x} = 0.696244$


Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.7$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.684454$; $x_0 \in [0, 1]$
$\bar{x} = 0.720043$; $x_0 \in [0, 1]$
$\bar{x} = 0.755493$; $x_0 \in [0, 1]$
$\bar{x} = 0.787227$; $x_0 \in (0.420713458, 1]$
$\rho = 0.6$, $\bar{x} = 0.684454$
$\rho = 0.7$, $\bar{x} = 0.720043$
$\rho = 0.8$, $\bar{x} = 0.755493$
$\rho = 0.9$, $\bar{x} = 0.787227$


Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.8$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.746980$; $x_0 \in [0, 1]$
$\bar{x} = 0.779998$; $x_0 \in [0, 1]$
$\bar{x} = 0.800000$; $x_0 \in [0, 1]$
$\bar{x} = 0.350000$; $x_0 \in [0, 0.530115762]$
$\bar{x} = 0.800000$; $x_0 \in (0.572249543, 1]$
$\rho = 0.6$, $\bar{x} = 0.746980$
$\rho = 0.7$, $\bar{x} = 0.779998$
$\rho = 0.8$, $\bar{x} = 0.800000$
$\rho = 0.9$, $\bar{x} = 0.350000$
$\rho = 0.9$, $\bar{x} = 0.800000$


Fluid competition: optimal policies and initial conditions | outside option $p_0 = 0.9$
Legend: persistence and equilibrium reputation; diamonds: stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
$\bar{x} = 0.800000$; $x_0 \in$ [0, 1]
$\bar{x} = 0.350000$; $x_0 \in$ [0, 0.464244581]
$\bar{x} = 0.800000$; $x_0 \in$ (0.485527169, 1]
$\bar{x} = 0.350000$; $x_0 \in$ [0, 0.64579659]
$\bar{x} = 0.800000$; $x_0 \in$ (0.664455417, 1]
$\bar{x} = 0.350000$; $x_0 \in$ [0, 1]
$\rho = 0.6$, $\bar{x} = 0.800000$
$\rho = 0.7$, $\bar{x} = 0.350000$
$\rho = 0.7$, $\bar{x} = 0.800000$
$\rho = 0.8$, $\bar{x} = 0.350000$
$\rho = 0.8$, $\bar{x} = 0.800000$
$\rho = 0.9$, $\bar{x} = 0.350000$